DATA AND INFORMATION MANAGEMENT

ISYS2120

Question Bank

Interactive practice and explained answers

150 tasks | 100 Canvas questions | 18 revision-paper tasks | 24 skill drills | 8-question practice exam

Open a section from Contents or the bookmark sidebar. Attempt a question, click View Answer, then use Back to Question.

Every visible completed Canvas question has an explicitly published answer marker. The recorded key is separated from independently written explanations and technical corrections.

Navigation matches the INFO3315 bank: clickable contents, question bookmarks and two-way question/answer links. Use PDF annotations to record your work.

Semester 2, 2026 | Personal study edition

Prepared from course PDFs and visible completed Canvas quiz reviews. Explanations and new exercises are independently written study support.

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Read this before practising

Question and answer conventions

Canvas option order and wording follow the visible completed submission. A-D labels are assigned in that order. Tables and five original ER images are included. Equations are transcribed using PROJECT, SELECT and JOIN with bracketed subscripts for legibility; the guide explains their symbolic equivalents. Source underlining is also recorded when it identifies keys.

The question pages do not reveal your prior selections. Answer pages show the published option, your visible selection, its score and the public answer marker, followed by an independently written explanation. "Published key" describes Canvas evidence; it does not mean every source item is technically flawless.

The ten quizzes cover Week 2, Week 3 Early Feedback, Weeks 5-8 and Weeks 10-13. No weekly quiz for Weeks 1, 4 or 9 was listed on the course Quizzes page. This captures 100 visible occurrences, including repeats, rather than claiming unseen question-pool coverage.

Source issues to learn from

Question IDs	Technical note
W03-Q02; W06-Q08	Unqualified SuppID is ambiguous in the supplied two-table SQL; the published key explains the intended missing-join effect.
W05-Q08	Catalog/Catalogue and color/colour are inconsistent between the source schema and options.
W08-Q06; W13-Q09	A distractor tuple contains a doubled comma. It is retained as printed.
W08-Q08	A source attribute list has B.C punctuation; the intended separator is explained.
W13-Q10	Canvas publishes A, but C also closes to the full relation. Both proofs appear in the answer.

The revision-paper prompts are reformatted for readability and keep their task requirements and suggested times. Their solutions and the 32 additional exercises are independently written, with no claim to be official marking guidance. The practice exam uses the 50-mark distribution reported in the supplied Week 13 review.

Your captured review set

68 of the visible selections match the published key; 32 differ. Repeated occurrences are counted separately, just as they appeared in their quizzes. This is review evidence, not a predicted exam result.

Review the 32 differing selections after your first attempt.

W02-Q01 | W02-Q06 | W02-Q07 | W02-Q10 | W03-Q06 | W03-Q08 | W05-Q02 | W05-Q04 | W05-Q08 | W05-Q09 | W06-Q02 | W06-Q06 | W06-Q07 | W06-Q09 | W06-Q10 | W07-Q03 | W07-Q05 | W07-Q06 | W07-Q09 | W08-Q03 | W08-Q05 | W10-Q02 | W10-Q04 | W10-Q09 | W10-Q10 | W11-Q01 | W11-Q08 | W11-Q10 | W12-Q03 | W12-Q06 | W12-Q10 | W13-Q02

Part I. Canvas weekly quizzes

100 questions from ten visible completed submissions. Try each question before following its answer link.

Week 02 - Weekly Quiz

week2 quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W02-Q01 | Question 1

View Answer

Which assessments in isys2120 are done as group work?

A. Assignment 2 (Concepts) only

B. Assignment 1 (Conceptual Model) and Assiignment 2 (Concepts)

C. There is no group-based assessment in isys2120

D. Assignment 1 (Conceptual Model) only

My choice: _____ Confidence: low / medium / high

W02-Q02 | Question 2

View Answer

What is the weighting in assessment for the Weekly Quizzes?

A. Each quiz is worth 1%; there are 9 quizzes, so the total from quizzes is 9%

B. Each quiz is worth 3%, there are 9 quizzes, so the total from quizzes is 27%

C. Each quiz is worth 3%; there are more than 7 quizzes but the total from quizzes is capped at 21%

D. Each quiz is worth 5%, there are 4 quizzes, so the total from quizzes is 20%

My choice: _____ Confidence: low / medium / high

W02-Q03 | Question 3

View Answer

Which of the following statements is true about lectures in isys2120

A. You may attend in person, or watch online at the same time, or watch the recording later

B. You must attend in person at least 80% of the lectures; attendance is recorded

C. Lectures are only available to watch online at the time they occur

D. You are required to watch the recording, obtained via Canvas

My choice: _____ Confidence: low / medium / high

W02-Q04 | Question 4

View Answer

Which of the following assessment tasks allows work to be submitted after the due date?

A. Assignment 1 (Conceptual Design)

B. Final Exam

C. Weekly Quiz

D. SQL Online Quiz

My choice: _____ Confidence: low / medium / high

W02-Q05 | Question 5

View Answer

How often are you expected to check your university email

A. Daily

B. at least once a week

C. you do not need to check email, as anything important will be posted on Ed

D. at least once per semester

My choice: _____ Confidence: low / medium / high

W02-Q06 | Question 6

View Answer

Which of the following is not an example of meta-data for a system that manages data about a companies employees

A. The legal obiigations regarding privacy of employee salaries

B. The fact that salary is kept for each employee

C. The salary for a particular employee

D. The source for information about employee salaries

My choice: _____ Confidence: low / medium / high

W02-Q07 | Question 7

View Answer

What of the following is not an activity which might be done by DBMS Vendor Staff?

A. Write the code of the DBMS

B. Train the organisations staff in the capabilities of the dbms platform

C. Write the code of an application used by endusers to produce a particular report

D. Write the code of a tool which can be used with the dbms, for tuning it

My choice: _____ Confidence: low / medium / high

W02-Q08 | Question 8

View Answer

What is a difference between a database, and a database management system

A. A database conatins the data used by a single application or user, while a database management system holds data that is shared between many applications or users

B. A database has only one table, while a database management system has many tables

C. A database is a collection of data, while a database management system is a software package through which many programs can access a shared database

D. There is no difference, the terms mean the same

My choice: _____ Confidence: low / medium / high

W02-Q09 | Question 9

View Answer

The term DML is used for

A. the language used to describe what tables exist and what structure they have

B. the logic that describes how to understand a command that defines the structure of a table

C. the logic that describes how to understand a command that access the data

D. the language used to access the data held in a dbms, either to update the data or to retrieve some data

My choice: _____ Confidence: low / medium / high

W02-Q10 | Question 10

View Answer

The language used to retrieve information from a typical relational dbms is

A. SQL

B. Python

C. English

D. PostgreSQL

My choice: _____ Confidence: low / medium / high

Week 03 - Early Feedback Quiz

Early Feedback Quiz (week 3)- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W03-Q01 | Question 1

View Answer

The weight of the early feedback quiz in isys2120 is

A. no weight; this is just for feedback on your progress

B. 3%; this cannot be made up using other weekly quizzes

C. 3%, but the best 7 quizzes are used, so a low score here will not matter if you do well on 7 others

D. no weight, but this is a hurdle for the unit; if you fail the EFQ, you fail the unit

My choice: _____ Confidence: low / medium / high

W03-Q02 | Question 2

View Answer

Consider two tables Supplier(SuppID, Name, Phone) and Product(ProdID, Descr, SuppID). What is an English language explanation of the output of the SQL query

```
SELECT ProdID, SuppID
FROM Supplier, Product
WHERE Name = 'Heinz'
```

A. Show every combination that pairs a SuppID from a row of the Supplier table with the ProdID of a row from the Product table

B. Show every combination that pairs a SuppID from a row of the Supplier table where the Name is 'Heinz', with the ProdID of a row from the Product table

C. Show all the information about the products which are supplied by Heinz.

D. Show the SuppID and ProdID for any produce which is supplied by Heinz

My choice: _____ Confidence: low / medium / high

W03-Q03 | Question 3

View Answer

Consider two tables Supplier(SuppID, Name, Phone) and Product(ProdID, Descr, SuppID). Which of the following does not produce as output, all the information in the Product table, for products whose supplier has ID 523?

A. SELECT P.ProdID, P.Descr, P.SuppID
FROM Product P
WHERE P.SuppID = 523

B. SELECT Descr
FROM Product
WHERE SuppID = 523

C. SELECT *
FROM Product
WHERE SuppID = 523

D. SELECT ProdID, Descr, SuppID
FROM Product
WHERE SuppID = 523

My choice: _____ Confidence: low / medium / high

W03-Q04 | Question 4

View Answer

Consider a table AssignmentMark(StudentId, UnitCode, AssignmentNumber, MaxPossible, Score). Which query finds the number of students for which there are marks in the database?

A. SELECT COUNT DISTINCT(StudentId)
FROM AssignmentMark;

B. SELECT COUNT(DISTINCT StudentID)
FROM AssignmentMark;

C. COUNT DISTINCT StudentID
FROM AssignmentMark;

D. DISTINCT COUNT (StudentID)
FROM AssignmentMark;

My choice: _____ Confidence: low / medium / high

W03-Q05 | Question 5

View Answer

In a diagram showing a relational schema, an underline on a collection of column names means

A. Every row of the table has a different combination of values in this set of columns

B. These columns have names that include "ID"

C. Every other column of the table repeats a value found in one of these columns

D. Each of these columns is of integer type

My choice: _____ Confidence: low / medium / high

W03-Q06 | Question 6

View Answer

Which of the following should not occur in an Entity-Relationship conceptual model

A. a relationship type with an attribute

B. an entity type with a multi-valued attribute

C. an entity type where the instances are unique based on a combination of several attributes, but no single attribute is able to distinguish all the instances

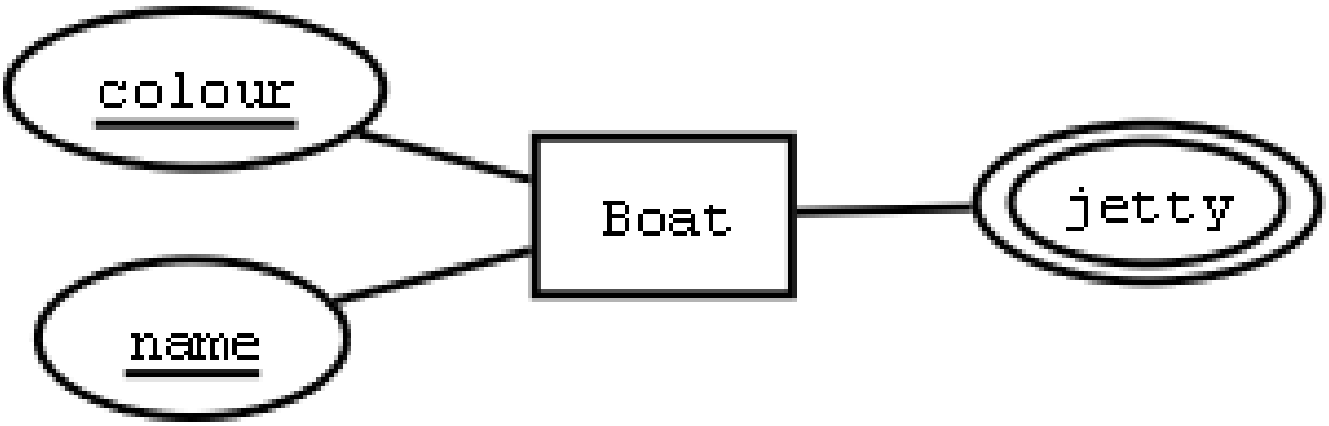
D. an entity type with an attribute which is the identifier of some other entity type

My choice: _____ Confidence: low / medium / high

W03-Q07 | Question 7

View Answer

Which is correct about the Boat Entity Type, based on the following ER diagram You can also download the image by the link
boat_entity.png



Original Canvas question image.

A. A boat has at most one jetty value.

B. A boat has exactly two jetty values.

C. A boat can have zero, one, or more jetty values

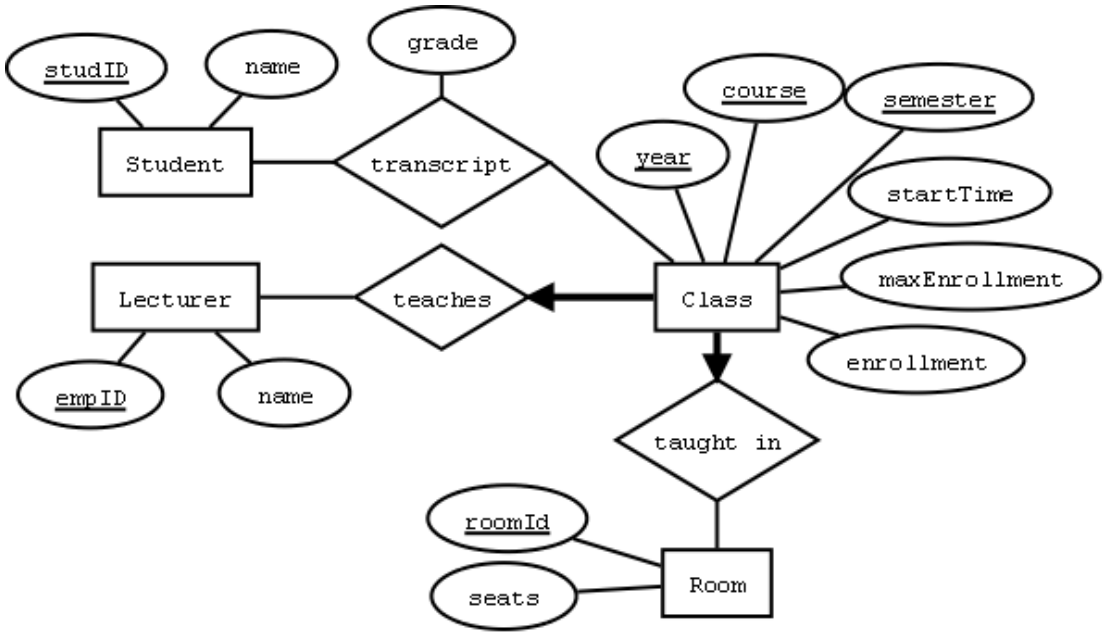
D. A boat has exactly one jetty value.

My choice: _____ Confidence: low / medium / high

W03-Q08 | Question 8

View Answer

Which one of the following statements is INCORRECT according to ER diagram below?



Original Canvas question image.

A. Whenever a lecturer can teaches several classes, they can't be in the same semester of the same year.

B. Each class is taught by exactly one lecturer.

C. There can be two classes which are from the same course, in the same year, but different semesters.

D. Several classes can be taught in the same room.

My choice: _____ Confidence: low / medium / high

W03-Q09 | Question 9

View Answer

Who is the unit coordinator for isys2120 in semester 2 of 2026?

A. Prof Wirth

B. Mr Rong

C. Dr Polash

D. Prof Fekete

My choice: _____ Confidence: low / medium / high

W03-Q10 | Question 10

View Answer

Which of the following is not a benefit from keeping data with a DBMS rather than in files that are directly accessed by application programs

A. a dbms offers organisational control of valuable data, while application programs might make choices that are not good for the overall needs

B. a dbms allows integrity constraints to be enforced across the data and changed as necessary, while in a file system, checking for integrity has to be done in each program's code separately (and so changes are difficult to manage)

C. a DBMS is cheap, while a file system is more expensive (both in fees for the software, and wages for expert staff)

D. a dbms offers a sophisticated security mechanism that can be adjusted for individual users and groups, and different parts of the database, while a file system usually has access control that is for the whole file at once, and with only a few categories of different levels of access

My choice: _____ Confidence: low / medium / high

Week 05 - Weekly Quiz

week 5 weekly quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W05-Q01 | Question 1

View Answer

Which of the following is correct SQL to define a table with two columns each holding strings

A. CREATE TABLE TableA (
column1 VARCHAR(4),
column2 VARCHAR(8)
);

B. CREATE TABLE TableA (
column1,
column2
);

C. CREATE TABLE TableA (
column1:VARCHAR(4),
column2: VARCHAR(8)
);

D. CREATE TABLE TableA (
VARCHAR(4) column1,
VARCHAR(8) column2
);

My choice: _____ Confidence: low / medium / high

W05-Q02 | Question 2

View Answer

Consider a database schema with the following two relations

AreaDepot(postcode: int, suburb: varchar(16), state:varchar(3), depot_name: varchar(20), lat: real, long:real)

Depot(name: varchar(20), manager: varchar(30))

The relation AreaDepot has an additional integrity constraint:

FOREIGN KEY(depot_name) REFERENCES Depot(name);

What is always true about an instance of the database?

Underlined in the source: suburb, state, name.

A. The set of values that appear in the name column of Depot is exactly the same as the set of values that appear depot_name column of AreaDepot.

B. Any non-NULL value in the depot_name column of AreaDepot, also occurs in the name column of Depot

C. Any value in the in the name column of Depot also occurs in the depot_name column of AreaDepot.

D. For any row of the AreaDepot table, the depot_name column contains a number that shows which row of the Depot table is involved.

My choice: _____ Confidence: low / medium / high

W05-Q03 | Question 3

View Answer

Suppose we have a relational table Supplier(SuppID, Name, Phone). Which command would include an extra row with contents (557, 'ACME', '0411203579')?

A. ADD
INTO Supplier
VALUES (557, 'ACME', '0411203579');

B. EXTEND Supplier
WITH (557, 'ACME', '0411203579');

C. UPDATE SUPPLIER
SET SuppID = 557, Name = 'ACME', Phone = '0411203579;

D. INSERT
INTO Supplier
VALUES (557, 'ACME', '0411203579');

My choice: _____ Confidence: low / medium / high

W05-Q04 | Question 4

View Answer

In a relational schema, we have a table TableT(col1, col2, col3, col4) where col1 is a foreign key referencing TableU, and col2 is a foreign key referencing TableV, and the primary key is composite (col1, col2). Which of the following could occur in an ER diagram that was mapped to this relational design?

A. A relationship type T between entities U and V, where T has attributes col1, col2, col3 and col4. The relationship T is many-to-many

B. A relationship type T between entities U and V, where T has attributes col1, col2, col3 and col4. The relationship T is many-to-one

C. A relationship type T between entities U and V, where T has attributes col3 and col4. The relationship T is many-to-one

D. A relationship type T between entities U and V, where T has attributes col3 and col4. The relationship T is many-to-many

My choice: _____ Confidence: low / medium / high

W05-Q05 | Question 5

View Answer

Consider the following instances for relation PostArea(postcode, suburb, state, depot_name)

postcode	suburb	state	depot_name
2528	BARRACK HEIGHTS	NSW	OAK FLATS DC
2528	BARRACK POINT	NSW	OAK FLATS DC
2535	BROGERS CREEK	NSW	NOWRA DC
3825	YALLOURN NORTH	VIC	MOE DELIVERY CENTRE
3831	NEERIM	VIC	NEERIM SOUTH

And Deliverer(postcode, emp_id, name)

postcode	emp_id	name
2528	101	Alice Walker
2535	102	Bob Strider
2528	103	Clare Runner
2535	103	Clare Runner
3831	104	David Stroller

Which of the following occurs as a row in the relation

(PostArea JOIN Deliverer)

Underlined in the source: suburb, state, postcode, emp_id.

A. 2528 BARRACK POINT NSW OAK FLATS DC 103 Clare Runner

B. 2528 BARRACK HEIGHTS NSW OAK FLATS DC 104 David Stoller

C. 2528 BROGERS CREEK NSW NOWRA DC 103 Clare Runner

D. 2528 BARRACK POINT NSW OAK FLATS DC 104 David Stoller

My choice: _____ Confidence: low / medium / high

W05-Q06 | Question 6

View Answer

For the relation

Car(regno, model, year, owner)

Which of the following relational algebra queries returns the registration numbers of all cars that were made before 2002?

Underlined in the source: regno.

A. SELECT[year<2002](PROJECT[regno](Car))

B. PROJECT[regno](SELECT[year<2002](Car))

C. SELECT[year>2002](Car)

D. SELECT[regno,year<2002](Car)

My choice: _____ Confidence: low / medium / high

W05-Q07 | Question 7

View Answer

Consider the following relations:
AcademicStaff(id, name, department, address, salary)
UoSOffering(uoSCode, semester, year, enrollment, instructorId), where instructorId is a foreign key that references AcademicStaff(id)

Which SQL query calculates the answer to "For each uoScode, show how many staff members have instructed the unit when there were at least 200 students enrolled"?

Underlined in the source: id, uoSCode.

A.
```
SELECT uoSCode, count(DISTINCT id)
    FROM UoSOffering JOIN AcademicStaff ON instructorId = id
    WHERE enrollment >= 200
    GROUP BY UoSOffering;
```

B.
```
SELECT uoSCode, count(DISTINCT id)
    FROM UoSOffering JOIN AcademicStaff ON instructorId = id
    GROUP BY id;
```

C.
```
SELECT uoSCode, count(DISTINCT id)
    FROM UoSOffering JOIN AcademicStaff ON instructorId = id
    WHERE enrollment >= 200
    GROUP BY uoSCode;
```

D.
```
SELECT uoSCode, count(DISTINCT id)
    FROM UoSOffering JOIN AcademicStaff ON instructorId = id
    HAVING enrollment >= 200
```

My choice: _____ Confidence: low / medium / high

W05-Q08 | Question 8

View Answer

Consider the following relations:

Supplier(sid, sname, address)

Part(pid, pname, color)

Catalog(sid, pid, cost)

The primary key fields are underlined. Therefore sid is the primary key for Suppliers, pid is the primary key for Parts, and sid and pidtogether form the primary key for Catalog. The Catalog relation lists the prices charged for parts by Suppliers.

Which of the following queries will find the ids of suppliers of red parts?

Underlined in the source: sid, pid, sid, pid.

A. SELECT c.sid
FROM Catalogue c
WHERE c.pid IN
(SELECT pid
FROM Part
WHERE colour = 'red');

B. SELECT c.sid
FROM Catalogue c
WHERE c.pid IS (SELECT pid
FROM Part
WHERE colour = 'red');

C. SELECT c.sid
FROM Catalogue c
WHERE c.pid = (SELECT pid
FROM Part)
AND colour = 'red');

D. SELECT sid
FROM Supplier
WHERE pid IN (SELECT pid
FROM Part
WHERE colour = 'red');

My choice: _____ Confidence: low / medium / high

W05-Q09 | Question 9

View Answer

What is the essential feature that indicates that there is a correlated subquery, instead of a non-correlated one?

A. The subquery involves aggregation over the rows of the table from the outer query

B. The subquery and the outer query have the same table in their FROM clauses

C. The subquery refer to one of the table variables from the outer query

D. The subquery is used in the WHERE clause of the outer query

My choice: _____ Confidence: low / medium / high

W05-Q10 | Question 10

View Answer

Consider the following relations:
AcademicStaff(id, name, department, address, salary)
UoSOffering(uoSCode, semester, year, enrollment, instructorId), where instructorId is a foreign key that references AcademicStaff(id)

What is being calculated by the query

```
SELECT a.name
FROM AcademicStaff a
WHERE a.department = 'SCS'
AND 5 <= (SELECT COUNT(Distinct u.uoSCode)
FROM UoSOffering u
WHERE u.instructorId = a.id
AND u.year = 2025);
```

Underlined in the source: id, uoSCode, semester, year.

A. find the name of academic staff in the "SCS" department, who taught at least 5 different UoS in 2025

B. find the name of academic staff who taught every unit in the 'SCS" department during 2025

C. find the name and uoSCode where the academic staff are in the "SCS" department and taught the unit at least 5 times in 2025

D. find the uoSCodes taught in the "SCS" department by at least 5 different academics in 2025

My choice: _____ Confidence: low / medium / high

Week 06 - Weekly Quiz

quiz-week6- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W06-Q01 | Question 1

View Answer

When analysing a text description of the domain, an entity type is often suggested by which category of word

A. preposition

B. noun

C. adverb

D. verb

My choice: _____ Confidence: low / medium / high

W06-Q02 | Question 2

View Answer

Which of the following aspects of text is not likely to provide a clue about a cardinality or participation constraint on a relationship type?

A. Using the word "must"

B. Using a verb that connects two nouns

C. Using a plural form of a noun

D. Using a singluar form of a noun

My choice: _____ Confidence: low / medium / high

W06-Q03 | Question 3

View Answer

Consider the following instance for relation Deliverer(postcode, emp_id, name)

postcode	emp_id	name
2528	101	Alice Walker
2535	102	Bob Strider
2528	103	Clare Runner
2535	103	Clare Runner
3831	104	David Stroller

What is the result of the operation to Project this on the emp_id attribute?

A. emp_id | 101 | 102 | 103 | 104

B. The project operator can't be applied to the Deliverer relation.

C. emp_id | 101 | 102 | 103 | 104 | 105

D. emp_id | 101 | 102 | 103 | 103 | 104

My choice: _____ Confidence: low / medium / high

W06-Q04 | Question 4

View Answer

What is the essential feature that indicates that there is a correlated subquery, instead of a non-correlated one?

A. The subquery involves aggregation over the rows of the table from the outer query

B. The subquery is used in the WHERE clause of the outer query

C. The subquery refer to one of the table variables from the outer query

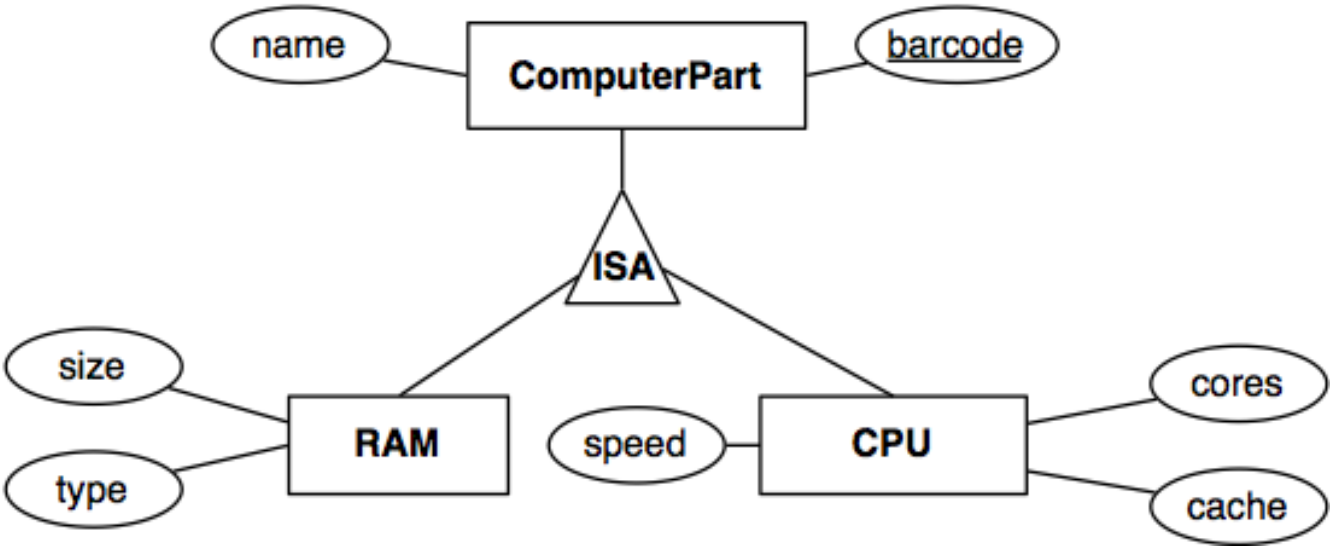
D. The subquery and the outer query have the same table in their FROM clauses

My choice: _____ Confidence: low / medium / high

W06-Q05 | Question 5

View Answer

We are considering part of the E/R diagram of a Computer Shop database, as shown below. What attributes would a CPU have?



Original Canvas question image.

A. size, type, cores, cache, speed

B. barcode, cores, cache, speed

C. cores, cache, speed

D. barcode, name, cores, cache, speed

My choice: _____ Confidence: low / medium / high

W06-Q06 | Question 6

View Answer

Which statement is correct about the following ER model.



Original Canvas question image.

A. A competitor must play more than one game

B. A competitor plays at most one game

C. A game is played by exactly one competitor

D. A game is played by at most one competitor

My choice: _____ Confidence: low / medium / high

W06-Q07 | Question 7

View Answer

Which of the following should not occur in an Entity-Relationship conceptual model

A. a relationship type with an attribute

B. an entity type with an attribute which is the identifier of some other entity type

C. an entity type where the instances are unique based on a combination of several attributes, but no single attribute is able to distinguish all the instances

D. an entity type with a multi-valued attribute

My choice: _____ Confidence: low / medium / high

W06-Q08 | Question 8

View Answer

Consider two tables Supplier(SuppID, Name, Phone) and Product(ProdID, Descr, SuppID). What is an English language explanation of the output of the SQL query

```
SELECT ProdID, SuppID
FROM Supplier, Product
WHERE Name = 'Heinz'
```

A. Show every combination that pairs a SuppID from a row of the Supplier table where the Name is 'Heinz', with the ProdID of a row from the Product table

B. Show all the information about the products which are supplied by Heinz.

C. Show the SuppID and ProdID for any produce which is supplied by Heinz

D. Show every combination that pairs a SuppID from a row of the Supplier table with the ProdID of a row from the Product table

My choice: _____ Confidence: low / medium / high

W06-Q09 | Question 9

View Answer

Consider the following relations:
AcademicStaff(id, name, department, address, salary)
UoSOffering(uoSCode, semester, year, enrollment, instructorId), where instructorId is a foreign key that references AcademicStaff(id)

Which SQL query calculates the answer to "For each uoSCode, show how many years it was offered in"

Underlined in the source: id, uoSCode, semester, year.

A. SELECT COUNT(DISTINCT year)

FROM UoSOffering

GROUP BY uoSCode

B. SELECT uoSCode, year

FROM UoSOffering

GROUP BY uoSCode

C. SELECT uoSCode, COUNT(DISTINCT year)

FROM UoSOffering

GROUP BY uoSCode

D. SELECT uoSCode

FROM UoSOffering

GROUP BY uoSCode

HAVING COUNT(DISTINCT year)

My choice: _____ Confidence: low / medium / high

W06-Q10 | Question 10

View Answer

Suppose we have a table Supplier(SuppID, Name, Phone). Which of the following commands will cause the table to become empty (have no rows)

Underlined in the source: SuppID.

A. DELETE *
IN Supplier;

B. DELETE
FROM Supplier;

C. DROP TABLE Supplier;

D. UPDATE Supplier
REMOVE *;

My choice: _____ Confidence: low / medium / high

Week 07 - Weekly Quiz

week7-weekly-quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W07-Q01 | Question 1

View Answer

Suppose that we have a relation R(U, V, W, X, Y) with the two functional dependencies which are UW -> X, Y -> V.

Suppose that R contains the tuple (5,6,7,8,9). Which of the following tuples can NOT also exist in R?

A. (43, 53, 7, 63, 73)

B. (30, 40, 50, 60, 70)

C. (5, 20, 7, 8, 9)

D. 85, 6, 75, 65, 9)

My choice: _____ Confidence: low / medium / high

W07-Q02 | Question 2

View Answer

This question is about a relational schema R(A,B,C,D,E,F,G,H,I) where there are six functional dependencies which are AC -> GI; B -> D; D -> C; E -> D; F ->E, G -> D.

Which of the following is the result of calculating the attribute closure {A,E}+

A. {A, B, C, D, E, F, G, I}

B. {A, C, D, E, G, I }

C. {A, E, D}

D. {A, C, D, E }

My choice: _____ Confidence: low / medium / high

W07-Q03 | Question 3

View Answer

This question is about a relational schema R(A,B,C,D,E,F,G,H,I) where there are six functional dependencies which are AC -> GI; B -> D; D -> C; E -> D; F ->E, G -> D.

Which of the following is also a functional dependency which necessarily holds for this schema?

A. CE -> A

B. CE -> F

C. CE -> GI

D. CE -> D

My choice: _____ Confidence: low / medium / high

W07-Q04 | Question 4

View Answer

This question is about a relational schema R(A,B,C,D,E,F,G,H,I) where there are six functional dependencies which are AC -> GI; B -> D; D -> C; E -> D; F ->E, G -> D.

Which of the following is a superkey for this schema?

A. {A, B, C, D, E, F, G}

B. {C, D, E, G, I}

C. {B, C, E, F, H}

D. {A, B, E, F, H}

My choice: _____ Confidence: low / medium / high

W07-Q05 | Question 5

View Answer

This question is about a relational schema R(A,B,C,D,E,F,G,H,I) where there are six functional dependencies which are AC -> GI; B -> D; D -> C; E -> D; F ->E, G -> D.

Which of the following is NOT a lossless-join decomposition for this schema?

A. R1(E, F) and R2(A, B, C, D, E, G, H, I)

B. R1(B,D) and R2(A, B, C, E, F, G, H, I)

C. R1(D, E) and R2(A, B, C, E, F, G, H, I)

D. R1(D, G) and R2(A, B, C, D, E, F, H, I)

My choice: _____ Confidence: low / medium / high

W07-Q06 | Question 6

View Answer

Consider a relation R, and a functional dependency P -> Q (where P and Q are sets of attributes of R).

Which of the following statements is a true statement

A. An instance of R cannot contain two tuples which have the same value for some attribute in Q, and different values for every attribute in P

B. An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

C. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and the same values for every attribute in P

D. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and different values for some attribute in P

My choice: _____ Confidence: low / medium / high

W07-Q07 | Question 7

View Answer

Given a relation R, with a set of functional dependencies F, which of the following calculations is a way to decide whether or not R is in BCNF

A. Consider every fd in F except those where the right hand side is a subset of the lefthand side; for each of these fds, check if the left hand side of that fd is a superkey of R. If this is the case for all of the considered fds, then we conclude that R is in BCNF, otherwise R is not in BCNF.

B. Consider every fd in F except those where the right hand side is a subset of the lefthand side; for each of these fds, check if the left hand side of that fd is a superkey of R. If this is the case for one of the considered fds, then we conclude that R is in BCNF, otherwise R is not in BCNF.

C. Consider every fd in F except those where the right hand side is a superset of the lefthand side; for each of these fds, check if the left hand side of that fd is a superkey of R. If this is the case for all of the considered fds, then we conclude that R is in BCNF, otherwise R is not in BCNF.

D. Consider every fd in F; for each of these fds, check if the right hand side of that fd is a superkey of R. If this is the case for one of the considered fds, then we conclude that R is in BCNF, otherwise R is not in BCNF.

My choice: _____ Confidence: low / medium / high

W07-Q08 | Question 8

View Answer

Suppose we have a relation R and a set F of functional dependencies on the relation.

Which of the following is not a true statement about any superkey of R?

A. if S is a superkey, then S+ contains all the columns of R

B. if S is a superkey, then R does not contain two rows that have the samne value for every attribute that is in S

C. if S is a superkey, then there exists a subset of S which is a candidate key fro R

D. if S is a superkey,then there exists an attribute A in S, such that removing A from S gives a candidate key for R is

My choice: _____ Confidence: low / medium / high

W07-Q09 | Question 9

View Answer

Consider a relation U(A, B, C, D, E). Suppose that we know that U is in BCNF, and that BD is a candidate key, AE is a candidate key, and no other candidate keys exist. Which of the following can be a functional dependency of the table U?

A. BD->AE

B. AB->DE

C. B->D

D. A->E

My choice: _____ Confidence: low / medium / high

W07-Q10 | Question 10

View Answer

Which of the following statements is always true about a lossless join decomposition of relation U into relations U1 and U2

A. When we join U1 with U2, we get a relation that contains all the tuples of the original relation U, and some extra tuples too.

B. When we join U1 with U2, we get the exactly the contents of the original relation U.

C. When we join U1 with U2, we get a relation that contains some but not all the tuples of the original relation U.

D. The functional dependencies in U1 and U2 imply exactly the fds of the original relation U.

My choice: _____ Confidence: low / medium / high

Week 08 - Weekly Quiz

week8-weekly-quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W08-Q01 | Question 1

View Answer

What is the SQL command to permit user X to change the value in column C of table T

A. ALLOW MODIFY(C) ON T TO X

B. GRANT UPDATE(C) ON T TO X

C. GRANT MODIFY(C) ON T TO X

D. ALLOW UPDATE(C) ON T TO X

My choice: _____ Confidence: low / medium / high

W08-Q02 | Question 2

View Answer

What is the effect when the owner of table T executes a SQL command

GRANT SELECT ON T TO X

WITH GRANT OPTION

A. X can execute SELECT statements that refer to table T, and X can also give SELECT access on T to other users

B. X can execute SELECT statements that refer to table T, and X can also give any privilege on T to other users

C. X can execute SELECT statements that refer to table T, and X can also give any privilege on any table to other users

D. X can execute SELECT statements that refer to table T, and X can also give SELECT access on any table to other users

My choice: _____ Confidence: low / medium / high

W08-Q03 | Question 3

View Answer

Which of the following in NOT a true statement about PostgreSQL roles

A. A user can hold at most 2 roles at any time

B. A role can be held by many users at the same time

C. The SQL command to remove a role from a user is REVOKE role FROM user

D. For each user, there is a role which is held only by that user

My choice: _____ Confidence: low / medium / high

W08-Q04 | Question 4

View Answer

Suppose we have a table R with columns A, B, C, D, and A is a primary key. What is a SQL command to make a view that contains just the values of B, each with the number of rows in R that contain that value for B

A. ON R VIEW VI(B, COUNT(A) GROUP BY B);

B. CREATE VIEW V1(B, Howmany) AS
SELECT B, COUNT(A)
FROM R
GROUP BY B;

C. GROUP VIEW V1 FROM R AS
B, COUNT(A);

D. VIEW V1(B, Howmany) ON R IS
B, COUNT(A)
GROUP BY B;

My choice: _____ Confidence: low / medium / high

W08-Q05 | Question 5

View Answer

Suppose we have a table of Employees(EmpID,Name, HomePhone, JobCategory, Salary). For which of the following access control policies can be done easily without creating a view

Underlined in the source: EmpID.

A. Permit a manager to see the salary of any employee whose salary is less than 25000, but not see the salary of other employees

B. Permit a manager to modify the Salary of any Employee, but not modify other information

C. Permit a manager to modify the salary of any employee in the "Secretarial" jobcategory, but not modify other information, nor to modify salaries of employees with other jobcategories.

D. Permit a manager to see the average salary for each jobcategory, but not see the salaries of individual employees

My choice: _____ Confidence: low / medium / high

W08-Q06 | Question 6

View Answer

Consider a relation R(A, B, C, D, E, F, G) with the following four functional dependencies:

CD -> BE
C -> G
E -> A
G -> C

Suppose that we know that R contains a tuple (3,4,5,6,7,8,9). Which of the following tuples is NOT also allowed to exist in R?

A. (3,31,5,33,34,35,9)

B. (62,63,5,65,66,8,9)

C. (44,4,5,6,48,49,9)

D. (3,,22,23,24,7,26,27)

My choice: _____ Confidence: low / medium / high

W08-Q07 | Question 7

View Answer

Consider a relation R(A, B, C, D, E, F, G) with the following four functional dependencies:

CD -> BE
C -> G
E -> A
G -> C

Which of the following is NOT a superkey for R?

A. {C,D,E,G}

B. {B,D,E,F,G}

C. {C,D,E,F}

D. {A,D,F,G}

My choice: _____ Confidence: low / medium / high

W08-Q08 | Question 8

View Answer

Consider a relation R(A, B, C, D, E, F, G) with the following four functional dependencies:

CD -> BE
C -> G
E -> A
G -> C

Which of the following is NOT a lossless-join decomposition of R?

A. R1(A,E) and R2(A,B.C,D,F,G)

B. R1(A,E) and R2(B,C,D,E,F,G)

C. R1(C,G) and R2(A,B,D,E,F,G)

D. R1(C,G) and R2(A,B,C,D,E,F)

My choice: _____ Confidence: low / medium / high

W08-Q09 | Question 9

View Answer

Consider a relation R, and a functional dependency P -> Q (where P and Q are sets of attributes of R).

Which of the following statements is a true statement

A. An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

B. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and different values for some attribute in P

C. An instance of R cannot contain two tuples which have the same value for some attribute in Q, and different values for every attribute in P

D. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and the same values for every attribute in P

My choice: _____ Confidence: low / medium / high

W08-Q10 | Question 10

View Answer

Which of the following statements is always true about a lossless join decomposition of relation U into relations U1 and U2

A. When we join U1 with U2, we get the exactly the contents of the original relation U.

B. When we join U1 with U2, we get a relation that contains all the tuples of the original relation U, and some extra tuples too.

C. When we join U1 with U2, we get a relation that contains some but not all the tuples of the original relation U.

D. The functional dependencies in U1 and U2 imply exactly the fds of the original relation U.

My choice: _____ Confidence: low / medium / high

Week 10 - Weekly Quiz

week10-quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W10-Q01 | Question 1

View Answer

What is the best description of a connection (also called "session"), for a database-backed application program?

A. A connection represents a pattern of a SQL statement, with placeholders that get filled in at run-time with specific values taken from host variables

B. A connection represents an ongoing communicationto the database, where setup occurs once (identifying the user, checking password, etc) and then several SQL statements can be performed one after another.

C. A connection represents a specific SQL statement, and allows this statement to be executed again and again

D. A connection represents the user, and allows them to perform SQL statements in many different databases

My choice: _____ Confidence: low / medium / high

W10-Q02 | Question 2

View Answer

How are the results returned to be displayed to the end-user of a database-backed web application

A. An HTML document is returned and rendered by the user's local browser

B. A Python dictionary is returnd which is processed by the end-users local Python code

C. A string is returned which is printed by the end-users local Python code

D. A SQL statement is returned and executed by the end-users local database

My choice: _____ Confidence: low / medium / high

W10-Q03 | Question 3

View Answer

What is a cursor used for, in database-backed application programming?

A. A cursor finds one row in a database table based on its primary key, without executing any SQL

B. A cursor represents a user who has been authenticated by checking the password

C. A cursor is a description of a family of queries, which differ in some specific values which are taken from user input

D. A cursor points to one row in the result set of a query, and allows the application to process one row after another

My choice: _____ Confidence: low / medium / high

W10-Q04 | Question 4

View Answer

In the code for week 8 lab, what constructs a query to send to the database

A. A emthod in the file database.py

B. A method in a file in the templates directory

C. A method in the file routes.py

D. A method in the file config.ini

My choice: _____ Confidence: low / medium / high

W10-Q05 | Question 5

View Answer

What is the SQL Injection security problem?

A. This is a software vulnerability that allows an attacker to submit some SQL fragment as user input which, through insufficiently-checked inclusion of this input within a query string, changes the meaning of a SQL statement that is executed in the dbms, in ways not intended by the application programmer.

B. It refers to incorrectly written software that exposes internal schema information through SQL error messages.

C. This is a security threat of gaining un-authorized access to a database system so that the attacker can send SQL statements directly to the database API through another user's login.

D. It refers to an application coding error where some input buffers have fixed lengths which are not checked when user input is written into it, allowing the attacker to change the code of the application software itself.

My choice: _____ Confidence: low / medium / high

W10-Q06 | Question 6

View Answer

Which statement is true about the storage technology provided by hard disks

A. each word can be accessed separately, just like with RAM (main memory),

B. data is lost in a power failure, just like with RAM (main memory)

C. storage is more expensive than RAM (main memory)

D. read or write of data is much slower than doing so in RAM (main memory)

My choice: _____ Confidence: low / medium / high

W10-Q07 | Question 7

View Answer

What is the role of the buffer pool in a database management platform

A. to ensure that data used is persistant, depsite crashes that can corrupt the secondary storage

B. to allow the database to store more data than will fit in the secondary storage

C. to prevent the database from being corrupted by application errors

D. to hide the slowness of access to secondary storage, by having needed data available in memory for when it will be accessed

My choice: _____ Confidence: low / medium / high

W10-Q08 | Question 8

View Answer

Which of the following statements is NOT true about a heap file with no index

A. inserting an extra record is fast

B. finding a record involves looking at each block in turn

C. the data is not arranged in an order based on the values

D. the data is always densely packed, without any gaps

My choice: _____ Confidence: low / medium / high

W10-Q09 | Question 9

View Answer

What is term used for an index which allows a query to be answered from the index itself without accessing the data records

A. secondary index for the query

B. covering index for the query

C. composite index for the query

D. clustered index for the query

My choice: _____ Confidence: low / medium / high

W10-Q10 | Question 10

View Answer

Suppose we have a table R with columns a, b, c where a is the primary key. Which of the following will cause all the records with the same b value to be next to one another in the data?

A. An unclustered index on (a,b)

B. A clustering index on b

C. A clustering index on (a,b)

D. An unclustered index on b

My choice: _____ Confidence: low / medium / high

Week 11 - Weekly Quiz

week11 weekly quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W11-Q01 | Question 1

View Answer

Which of the following is NOT true about a stored procedure

A. The stored procedure is executed outside the dbms, passing SQL queries to the dbms and receiving the results

B. The stored procedure can be invoked with a CALL statement

C. The stored procedure is written in a language that varies from dbms to dbms, and combines SQL with features like loops and local variable

D. The stored procedure is run inside the dbms, and passes a whole result set back to the requesting client

My choice: _____ Confidence: low / medium / high

W11-Q02 | Question 2

View Answer

Which of the following is NOT true about an application that uses a call level interface for accessing a database?

A. The application is a mixture of normal language features, and special syntax for SQL statements

B. The application is written in a programming language, and processed with the normal compiler/interpreter for that language

C. The application uses strings that contain SQL text, and passes them as arguments in functions or methods.

D. The application invokes code in a library that provides common methods, but is implemented using a driver that is specific for a given dbms platform

My choice: _____ Confidence: low / medium / high

W11-Q03 | Question 3

View Answer

In the code for week 8 lab, what constructs a query to send to the database

A. A method in a file in the templates directory

B. A method in the file database.py

C. A method in the file config.ini

D. A method in the file routes.py

My choice: _____ Confidence: low / medium / high

W11-Q04 | Question 4

View Answer

What is term used for an index which allows a query to be answered from the index itself without accessing the data records

A. covering index for the query

B. secondary index for the query

C. composite index for the query

D. clustered index for the query

My choice: _____ Confidence: low / medium / high

W11-Q05 | Question 5

View Answer

Suppose we have a table R with columns a, b, c where a is the primary key. Which of the following will cause all the records with the same b value to be next to one another in the data?

A. An unclustered index on (a,b)

B. An unclustered index on b

C. A clustering index on (a,b)

D. A clustering index on b

My choice: _____ Confidence: low / medium / high

W11-Q06 | Question 6

View Answer

Which of the foillowing statements is correct about a table scan

A. A table scan is usually very efficient, because it directly finds the records that are needed without examining any others

B. A table scan cannot be used for a query that contains a WHERE clause

C. A table scan examines each block of data in the table, and within each block, it checks each record

D. A table scan can only be used in a secondary index exists on the table

My choice: _____ Confidence: low / medium / high

W11-Q07 | Question 7

View Answer

Which of the following statements is NOT correct about database tuning?

A. database tuning is done by the system when processing an SQL query, in order to make sure the query gives correct results

B. database tuning considers the tradeoffs in impact of update statements, whend eciding whether to create an extra index

C. database tuning aims to speed up the performance of a workload of queries

D. database tuning considers the queries that make up the workload, and then can suggest which indices to create

My choice: _____ Confidence: low / medium / high

W11-Q08 | Question 8

View Answer

Which of the following statements is most appropriate about query optimisation?

A. In query optimisation, the system executes the same SQL query several times, and then chooses the results if they all agree

B. In query optimisation, the system considers several different ways to execute a particular query, and chooses one which hopefully will run quickly

C. In query optimisation, the system considers several different SQL statements, and then chooses one which hopefully will give the most accurate results considers

D. In query optimisation, the system considers several possible indices it might create, and then creates the one that will use least space

My choice: _____ Confidence: low / medium / high

W11-Q09 | Question 9

View Answer

Consider a table T(a,b,c,d) with a as PRIMARY KEY. If we have created a clustering, hash-based secondary index on d, which of the following queries can be processed effectively using the secondary index?

A. SELECT d
FROM T
WHERE a = 32;

B. SELECT *
FROM T
WHERE 12 < c AND c < 42 AND d = 26;

C. SELECT *
FROM T
WHERE a = 32 AND 12 < c AND c < 42;

D. SELECT *
FROM T
WHERE c = 12 AND d > 26 AND d < 53;

My choice: _____ Confidence: low / medium / high

W11-Q10 | Question 10

View Answer

Suposse we have a table T(a,b,c,d) with a as Primary Key, and a composite tree-based unclusetred index on (d,c).

Which of the foillowing queries can make good use of the index on (d,c)

A. SELECT c FROM T
WHERE a = 3 AND b = 7;

B. SELECT a FROM T
WHERE b = 7 AND d = 4;

C. SELECT d FROM T
WHERE a = 3 AND b = 7;

D. SELECT a FROM T
WHERE b = 7 AND c = 6;

My choice: _____ Confidence: low / medium / high

Week 12 - Weekly Quiz

week12-quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W12-Q01 | Question 1

View Answer

In the ACID acronym, I stands for

A. Integrity

B. Independence

C. Isolation

D. Imperative

My choice: _____ Confidence: low / medium / high

W12-Q02 | Question 2

View Answer

A transaction is supposed to be "atomic". This means

A. the transaction keeps the database intergity constraints valid

B. either all the changes made by the transaction are present in the database, or none of the changes are present

C. the transaction has executed up to the final statement

D. the transaction contains several different SQL commands

My choice: _____ Confidence: low / medium / high

W12-Q03 | Question 3

View Answer

A transaction is supposed to be "consistent". Which of the following is NOT a true statement about a consistent transaction

A. every integrity constraint holds at every state of the database during execution of the transaction code

B. ensuring that a transaction is consistent is an obligation of the programmer who writes the transaction

C. the transaction must either commit or rollback

D. if the transaction starts when the database state satisfies every integrity constraint, then at the end of the transaction, the database state satisfies every integrity constraint

My choice: _____ Confidence: low / medium / high

W12-Q04 | Question 4

View Answer

Which of the following statements is not true about transaction isolation levels?

A. SQL allows the application program to tell the system which isolation level to provide for a transaction, through the SET TRANSACTION ISOLATION LEVEL statement

B. the SQL standard requires that the default isolation level is serializable (and so a system would ensure that undeclared integrity constraints hold as long as each application is written to be consistent)

C. many dbms platforms have default isolation that is weaker than serializable (and so they do not ensure that undeclared integrity constraints hold even when each application is written to be consistent)

D. system performance is usually unaffected by the choice of isolation level

My choice: _____ Confidence: low / medium / high

W12-Q05 | Question 5

View Answer

In an execution of serializable transactions

A. the application programmer ensures that the outcome is the same as in some execution where the transactions are run one after the other

B. the system ensures that each transactions sees no changes made by any other transaction

C. the system ensures that the outcome is the same as in some execution where the transactions are run one after the other

D. the application programmer ensures that each transactions sees no changes made by any other transaction

My choice: _____ Confidence: low / medium / high

W12-Q06 | Question 6

View Answer

Which statement is correct about the following ER model.



Original Canvas question image.

A. A competitor plays at most one game

B. A competitor must play more than one game

C. A game is played by at most one competitor

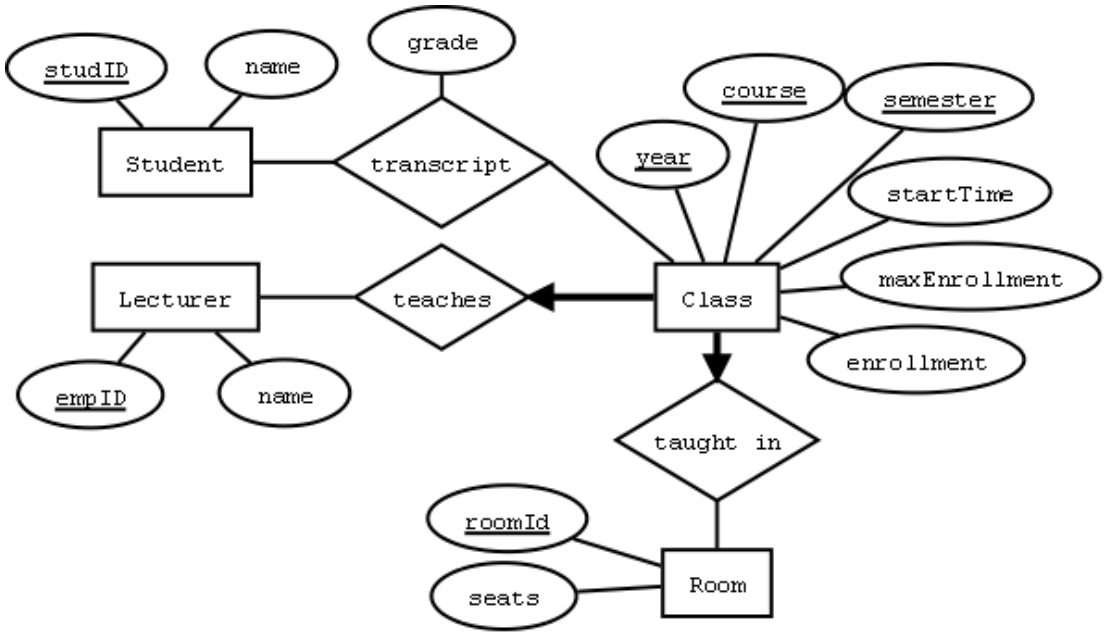
D. A game is played by exactly one competitor

My choice: _____ Confidence: low / medium / high

W12-Q07 | Question 7

View Answer

Which one of the following statements is INCORRECT according to ER diagram below?



Original Canvas question image.

A. Whenever a lecturer can teaches several classes, they can't be in the same semester of the same year.

B. There can be two classes which are from the same course, in the same year, but different semesters.

C. Each class is taught by exactly one lecturer.

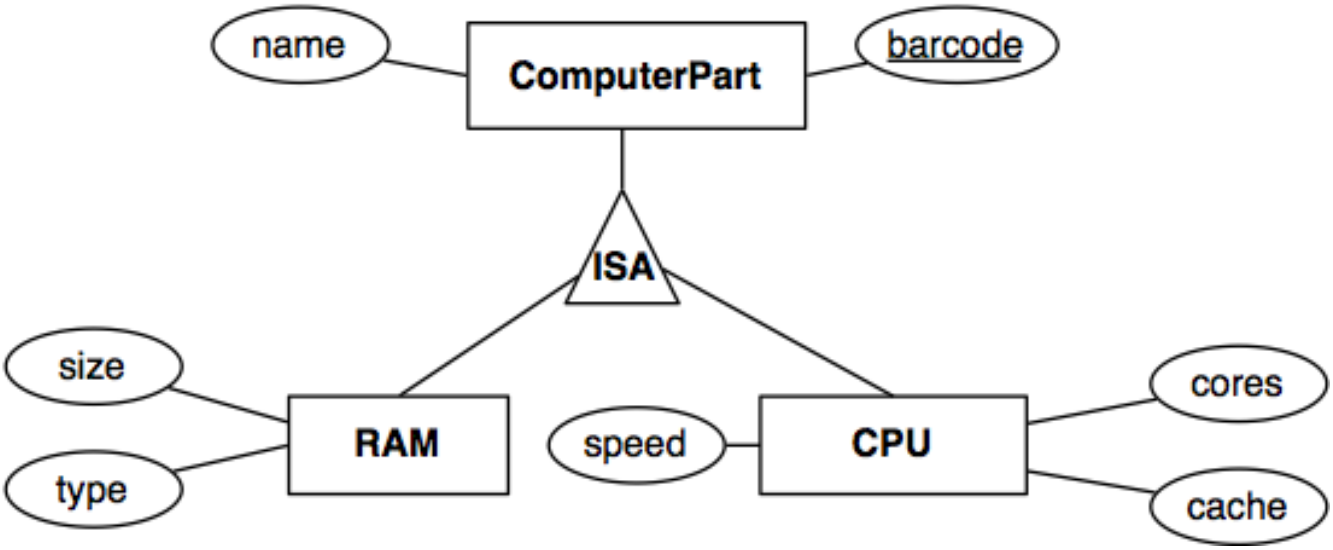
D. Several classes can be taught in the same room.

My choice: _____ Confidence: low / medium / high

W12-Q08 | Question 8

View Answer

We are considering part of the E/R diagram of a Computer Shop database, as shown below. What attributes would a CPU have?



Original Canvas question image.

A. size, type, cores, cache, speed

B. barcode, cores, cache, speed

C. cores, cache, speed

D. barcode, name, cores, cache, speed

My choice: _____ Confidence: low / medium / high

W12-Q09 | Question 9

View Answer

Suppose an ER diagram contains a weak entity type W, with a discriminator (partial key) attr1, and other attributes attr2 and attr3; the weak entity is dependent on an owning entity type S with primary key attrS. Which of the following is a suitable relational schema to capture this information?

A. A table tableW(attrS, attr1, attr2, attr3) where the primary key is attrS

B. A table tableW(attr1, attr2, attr3) where the primary key is attr1

C. A table tableW(attrS, attr1, attr2, attr3) where the primary key is (attrS, attr1) and attrS is a foreign key referencing tableS

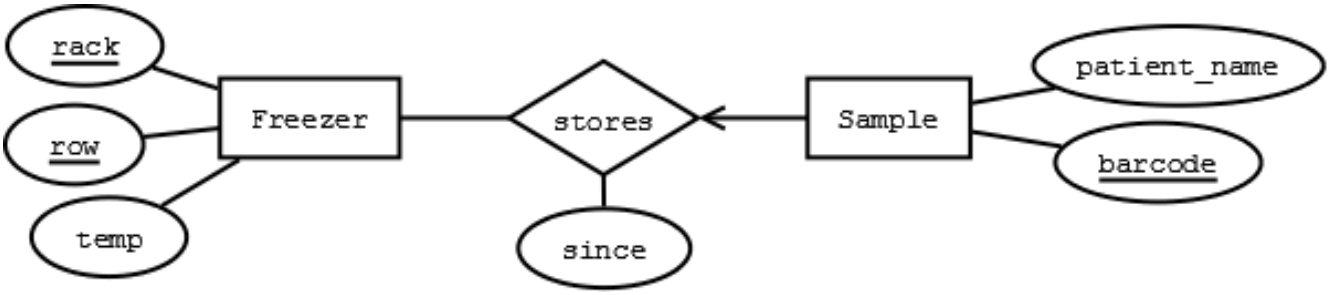
D. A table tableW(attrS, attr1) where the primary key is (attrS, attr1) and attrS is a foreign key referencing tableS

My choice: _____ Confidence: low / medium / high

W12-Q10 | Question 10

View Answer

The following ER model describes the storage of samples (blood, saliva, urine, etc.) collected from patients a hospital's pathology lab. Which relational schema best captures the information, including constraints? The image can also be downloaded: 1 sample_storage_ERD.png



Original Canvas question image.

A. Freezer(rack, row, temp);

Sample(barcode, patient_name);

Stores(since);

Underlined in the source: rack, row, barcode.

B. Freezer(rack, row, temp);

Sample(barcode, patient_name, freezer_rack, freezer_row, stored_since);

Constraints:

Foreign key Sample(freezer_rack, freezer_row) references Freezer(rack, row)

Underlined in the source: rack, row, barcode.

C. Freezer(rack, row, temp);

Sample(barcode, patient_name);

Stores(sample_barcode, freezer_rack, freezer_row);

Constraints:

Foreign key Stores(sample_barcode) references Sample(barcode)
Foreign key Stores(freezer_rack, freezer_row) references Freezer(rack, row)

Underlined in the source: rack, row, barcode, sample_barcode, freezer_rack, freezer_row.

D. Freezer(rack, row, temp, sample_barcode, stored_since);

Sample(barcode, patient_name);

Constraints:

Foreign key Freezer(sample_barcode) references Sample(barcode)

Underlined in the source: rack, row, barcode.

My choice: _____ Confidence: low / medium / high

Week 13 - Weekly Quiz

week13-quiz- Requires Respondus LockDown Browser

10 single-choice questions | Visible attempt 1 | Captured 14 November 2026

W13-Q01 | Question 1

View Answer

A difference between the queries in an OLAP system, and queries in an OLTP system, is:

A. OLAP queries are usually simple, while OLTP queries are usually complex

B. OLAP queries often look at a large part of the database, while OLTP queries usually deal with only a few items

C. OLAP queries often update the data, while OLTP queries rarely do updating

D. OLAP queries require up-to-date data, while OLTP queries can run with data that is a few days old

My choice: _____ Confidence: low / medium / high

W13-Q02 | Question 2

View Answer

In a ROLAP system, the information in the data warehouse

A. is stored in formats such as text, images etc, and accessed by programs in Python and other high-level languages

B. is stored in a multi-dimensional ("cube") structure and accessed by complex SQL queries

C. is stored in the contents of tables, and accessed through SQL

D. is stored in separate databases, and accessed by special ETL tools

My choice: _____ Confidence: low / medium / high

W13-Q03 | Question 3

View Answer

Which is a feature common in data warehouses:

A. A data warehouse is accessed by experts who write SQL queries interactively

B. a data warehouse is updated continually

C. A data warehouse stores data extracted from many operational databases

D. A data warehouse has a schema that is identical to that of the OLTP data sources

My choice: _____ Confidence: low / medium / high

W13-Q04 | Question 4

View Answer

The CUBE operation in SQL indicates that

A. the output is presented as a data cube rather than as a table

B. the running time is the cube of the number of rows in the data

C. "marginal" summary data is included in the output as well as the usual grouped aggregates

D. the output has "marginal" summary data, instead of the usual grouped aggregates

My choice: _____ Confidence: low / medium / high

W13-Q05 | Question 5

View Answer

A fact table gives the values for one or more measures, which depend on values for other attributes (such as location, time, etc). These other attributes are called

A. documents

B. dimensions

C. discussions

D. deviations

My choice: _____ Confidence: low / medium / high

W13-Q06 | Question 6

View Answer

Consider a relation PostArea(postcode, suburb, state, depot_name). What is the meaning of the relational algebra operation

SELECT[state='NSW'](PostArea)

Underlined in the source: suburb, state.

A. The subset of tuples from PostArea, consisting of the tuples for which state attribute has value 'NSW'

B. The values from the state column of the PostArea relation, except for those which are 'NSW'

C. The subset of tuples from PostArea, consisting of the tuples, leaving out those for which state attribute has value 'NSW'

D. The subset of suburb values from the PostArea relation, from tuples for which state attribute has the value 'NSW'

My choice: _____ Confidence: low / medium / high

W13-Q07 | Question 7

View Answer

Consider a schema Person(license, name, address) and Car(regno, model, year, owner), where owner is a foreign key that references Person(license). Which of the following relational algebra expressions calculates the name of the owner of the car whose regno is NSW314?

Underlined in the source: license, regno.

A. PROJECT[name](SELECT[regno = 'NSW314'](Person JOIN[owner = license] Car))

B. SELECT[regno = 'NSW314'](Car JOIN Person)

C. Person JOIN (PROJECT[regno = 'NSW314'](Car))

D. SELECT[owner=license,'NSW314',](PROJECT[name](Person JOIN Car))

My choice: _____ Confidence: low / medium / high

W13-Q08 | Question 8

View Answer

Consider a relation R, and a functional dependency P -> Q (where P and Q are sets of attributes of R).

Which of the following statements is a true statement

A. An instance of R cannot contain two tuples which have the same value for some attribute in Q, and different values for every attribute in P

B. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and different values for some attribute in P

C. An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

D. An instance of R cannot contain two tuples which have the same value for each attribute in Q, and the same values for every attribute in P

My choice: _____ Confidence: low / medium / high

W13-Q09 | Question 9

View Answer

Consider a relation R(A, B, C, D, E, F, G) with the following four functional dependencies:

CD -> BE
C -> G
E -> A
G -> C

Suppose that we know that R contains a tuple (3,4,5,6,7,8,9). Which of the following tuples is NOT also allowed to exist in R?

A. (62,63,5,65,66,8,9)

B. (3,,22,23,24,7,26,27)

C. (3,31,5,33,34,35,9)

D. (44,4,5,6,48,49,9)

My choice: _____ Confidence: low / medium / high

W13-Q10 | Question 10

View Answer

Consider a relation R(A, B, C, D, E, F, G) with the following four functional dependencies:

CD -> BE
C -> G
E -> A
G -> C

Which of the following is a superkey for R?

A. {C,D,F,G}

B. {A,B,C,E,G}

C. {D,E,F,G}

D. {C,D,E,G}

My choice: _____ Confidence: low / medium / high

Part II. Supplied revision-paper tasks

18 subquestions from the supplied five-page revision paper. Prompts are lightly reformatted; contexts and diagram are preserved. Suggested source times are retained.

Revision paper A - SQL and access

Member(MemberID, Name, Joined, Role, Location), PK(MemberID).
Activity(ActivityID, ActivityType, Leader, Duration), PK(ActivityID), FK Leader -> Member(MemberID). ActivityType and Duration are NOT NULL; Leader is nullable.
WorksOn(MemberID, ActivityID, NumberDaysWorked), PK(MemberID,ActivityID), with FKs to Member and Activity. Name and Joined in Member are NOT NULL.
Minor source punctuation is normalised. The source query's Members is treated as the declared table Member.

R-A1

View Answer

[4 minutes] Count activities whose duration is more than two days and whose leader has role 'Tester'.

Shared schema / diagram

Supplied revision paper p.1, A(i)

R-A2

View Answer

[4 minutes] Report, for each activity, how many members worked on it for more than three days.

Shared schema / diagram

Supplied revision paper p.1, A(ii)

R-A3

View Answer

[5 minutes] Calculate total hours worked on activity 57, assuming eight hours per work day. Two and three days contributed by two members should give 40 hours.

Shared schema / diagram

Supplied revision paper p.1, A(iii)

R-A4

View Answer

[6 minutes] Return all ActivityIDs of Testing activities with the greatest duration among Testing activities. Retain ties and order numerically by ActivityID.

Shared schema / diagram

Supplied revision paper pp.1-2, A(iv)

R-A5

View Answer

[5 minutes] Explain in English what restrictions Activity.Leader referencing Member imposes on table contents.

Shared schema / diagram

Supplied revision paper p.2, A(v)

R-A6

View Answer

[5 minutes] Give algebra returning names of members who worked more than three days on an activity of type Testing.

Shared schema / diagram

Supplied revision paper p.2, A(vi)

R-A7

View Answer

[5 minutes] Jane owns the schema. Give commands so Joe can see only names and roles of members working on activity 39 and cannot modify information.

Shared schema / diagram

Supplied revision paper p.2, A(vii)

R-A8

View Answer

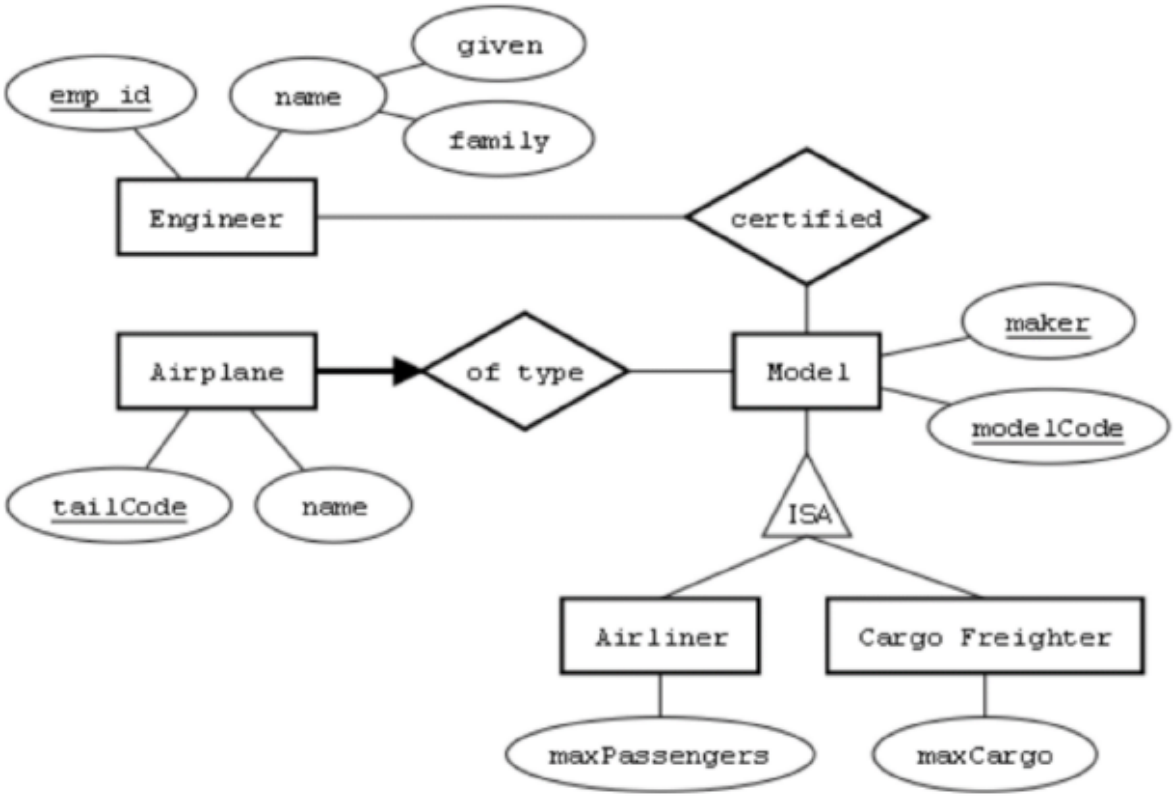
[7 minutes] Create and explain an index for SELECT MemberID, Name FROM Members WHERE Joined < '2022-01-20' AND Role = 'Tester' on a large table.

Shared schema / diagram

Supplied revision paper p.2, A(viii)

Revision paper B - Airport EER

Use the original airport EER diagram below. modelCode identifies Model; tailCode identifies Airplane; emp_id identifies Engineer. Arrow direction, thickness and the unannotated ISA hierarchy follow the unit notation.



Original airport diagram from the supplied revision paper, PDF p.3.

R-B1

View Answer

[3 minutes] Explain the connected attribute ellipses name, given and family.

Shared schema / diagram

Supplied revision paper p.3, B(i)

R-B2

View Answer

[4 minutes] Can airplane QXS213, named Spirit of Sydney, use model 373 made by Boeing while airplane VGY961, named HappyDays, uses model 373 made by Airbus? Explain using the diagram.

Shared schema / diagram

Supplied revision paper p.3, B(ii)

R-B3

View Answer

[12 minutes] Give CREATE TABLE statements mapping the airport EER diagram and its constraints where possible.

Shared schema / diagram

Supplied revision paper p.3, B(iii)

Revision paper C - DoctorsInfo

DoctorsInfo(VisitID, VisitDate, PatientID, Age, City, PostCode, ProvidorNo, Specialty, Diagnosis).
FDs: PatientID -> Age,PostCode,City; PostCode -> City; ProvidorNo -> Specialty; VisitID -> VisitDate,PatientID.
The sample includes:
(118,'Jan 3',318,22,'Sydney',2000,329,'Cardio','Murmur')
(118,'Jan 3',318,22,'Sydney',2000,295,'Lung','TB')
(129,'Jul 2',452,17,'Canberra',2600,329,'Cardio','Angina')
(213,'Jul 2',318,22,'Sydney',2000,183,'Lung','Pneumonia')
(234,'Sep 2',591,17,'Brisbane',4000,329,'Cardio','Healthy')
Only the listed FDs may be assumed. None determines Diagnosis.

R-C1

View Answer

[4 minutes] Which FD is contradicted by inserting (234,'Sep 2',591,17,'Brisbane',4000,183,'ENT','Cough')? Explain.

Shared schema / diagram

Supplied revision paper p.4, C(i)

R-C2

View Answer

[4 minutes] Calculate (VisitID,ProvidorNo)+ step by step.

Shared schema / diagram

Supplied revision paper p.4, C(ii)

R-C3

View Answer

[4 minutes] Justify why DoctorsInfo is not BCNF.

Shared schema / diagram

Supplied revision paper p.4, C(iii)

R-C4

View Answer

[8 minutes] Give a binary lossless decomposition. State each relation's FDs, a primary key and BCNF status.

Shared schema / diagram

Supplied revision paper p.4, C(iv)

R-C5

View Answer

[3 minutes] Prove the lossless-join property of the decomposition you gave.

Shared schema / diagram

Supplied revision paper p.4, C(v)

R-C6

View Answer

[3 minutes] Is your decomposition dependency-preserving? Justify.

Shared schema / diagram

Supplied revision paper p.5, C(vi)

Revision paper D - Flask security

R-D1

View Answer

[12 minutes] Teach a new developer who knows Python and SQL, but not Flask, how to inspect an extension of the Week 8 scaffold for SQL injection. Be explicit about files and functions.

Supplied revision paper p.5, D and local Week 8 ZIP

Part III. Additional skill drills

24 original short tasks, two for each broad learning block. Explain your reasoning in addition to the final answer.

Additional skill drills

D01

View Answer

Two Student rows are inserted, then an index is created on country. Distinguish instance, logical schema and physical design changes.

Original practice - Foundations

D02

View Answer

Two applications keep separate employee addresses. Explain one benefit and one cost of shared DBMS-managed data.

Original practice - Foundations

D03

View Answer

Each Chef prepares at most one Dish; every Dish must have at least one Chef. Specify both edges in the unit's notation.

Original practice - EER

D04

View Answer

Product has barcode and name. CPU adds cores and speed; RAM adds size. Covering is partial and overlap is allowed. List a CPU's attributes and explain those constraints.

Original practice - EER

D05

View Answer

Building's key is buildingID. Weak Room has roomNo unique only within Building and has capacity. Map Room and state what its owner FK does not enforce.

Original practice - Mapping

D06

View Answer

Employee may have at most one ParkingSpace; each space may have at most one employee. Both sides are optional. Give one FK mapping.

Original practice - Mapping

D07

View Answer

Activity IDs are 1 and 2. WorksOn contains (10,1,2 days) and (11,1,5 days). Count members contributing more than three days for every activity.

Original practice - SQL

D08

View Answer

Scores are 10,NULL,20,10. Calculate COUNT(*), COUNT(score), COUNT(DISTINCT score), SUM(score) and AVG(score).

Original practice - SQL

D09

View Answer

Student has IDs 1 and 2; Enrolled has only (1,'X'). RequiredUnit is empty. Who satisfies the double NOT EXISTS query over Student for having taken every required unit?

Original practice - Subqueries

D10

View Answer

A nullable-FK subquery returns {1,NULL}. Why can NOT IN fail to select unmatched candidate 2? Give an alternative.

Original practice - Subqueries

D11

View Answer

For Car(regno,year,owner), why is SELECT_year<2002(PROJECT_regno(Car)) invalid? Correct it.

Original practice - Algebra

D12

View Answer

Suppliers(sid,name), Parts(pid,colour), Catalog(sid,pid). Describe algebra for names of suppliers supplying both a red and a green part.

Original practice - Algebra

D13

View Answer

For R(A,B,C,D), F={A->B,B->C}, find all candidate keys and test BCNF.

Original practice - Design theory

D14

View Answer

For R(A,B,C), F={A->B}, compare decompositions AB/AC and AB/BC for losslessness.

Original practice - Design theory

D15

View Answer

An analyst may see average salary per department but no individual salaries. Propose an interface and a remaining disclosure concern.

Original practice - Security

D16

View Answer

X gets SELECT ON T WITH GRANT OPTION and grants it to Y. The owner also grants SELECT directly to Y. What matters when X's path is revoked?

Original practice - Security

D17

View Answer

A helper binds a search term but interpolates request.form['column'] as an SQL identifier. Is value binding enough?

Original practice - Applications

D18

View Answer

Contrast stored XSS and SQL injection. Why does one defence not automatically stop both?

Original practice - Applications

D19

View Answer

Compare tree indexes (role,joined) and (joined,role) for role='Tester' AND joined<'2022-01-20'.

Original practice - Indexes

D20

View Answer

A query needs records scattered over 900 of 1,000 table pages through an unclustered index. Why might a scan win?

Original practice - Indexes

D21

View Answer

A task commits a debit, then begins a separate credit transaction. Identify the ACID issue if the credit fails.

Original practice - Transactions

D22

View Answer

Two doctors each see the other on call and concurrently turn their own row off. Why is avoiding same-row lost updates insufficient?

Original practice - Transactions

D23

View Answer

For market and product, list CUBE's conceptual groupings and distinguish measures from dimensions.

Original practice - Analytics

D24

View Answer

A dataset is downloadable. Why does that alone not establish authority to reuse Indigenous knowledge in it?

Original practice - Governance

Part IV. Practice exam - 50 marks

Independent practice paper. Allow 10 minutes to read, then 120 minutes to write: about 100 minutes for a first pass and 20 for checking. The suggested marks total 50; they are a study rubric, not an official marking scheme.

Attempt every question before opening its solution. Use the ISYS2120 diagram notation.

Practice exam - 50 marks

M01 | 8 marks

View Answer

A conference records Workshop(workshopID,name), Room identified by (floor,roomNo) with capacity, and Participant identified by email with name. Each workshop uses exactly one room for all its dates. Each workshop has one or more day records, with a date and the day's chairperson name. Every participant registers for at least one workshop. Design an EER model in ISYS2120 notation; state assumptions.

Original practice based on course concepts

Member(MemberID, Name, Joined, Role, Location), PK(MemberID).
Activity(ActivityID, ActivityType, Leader, Duration), PK(ActivityID), FK Leader -> Member(MemberID). ActivityType and Duration are NOT NULL; Leader is nullable.
WorksOn(MemberID, ActivityID, NumberDaysWorked), PK(MemberID,ActivityID), with FKs to Member and Activity. Name and Joined in Member are NOT NULL.
Minor source punctuation is normalised. The source query's Members is treated as the declared table Member.

M02 | 6 marks

View Answer

Use the activity schema below. (a) Report every ActivityID and the number of members with NumberDaysWorked>3, including zeros. (b) Return all longest Testing ActivityIDs, retaining ties and sorting by ID.

Original practice based on course concepts

M03 | 8 marks

View Answer

Building(buildingID,name) owns weak Room, discriminator roomNo, attribute capacity. Booking(bookingID,startTime) is made by exactly one Customer(customerID,name) and uses exactly one Room. Each customer and room may have many bookings, including none. Map to tables, keys, FKs and nullability. Do not invent a no-overlap rule.

Original practice based on course concepts

M04 | 6 marks

View Answer

(a) For Person(license,name) and Car(regno,owner,year), give algebra for names of owners of pre-2002 cars, matching owner=license. (b) Explain covering versus clustered indexes. (c) Does an index on (a,b) make all equal-b records adjacent?

Original practice based on course concepts

M05 | 4 marks

View Answer

For R(A,B,C,D,E), F={A->BC,C->D,DE->A}, compute (CE)+ and find all candidate keys with a minimality argument.

Original practice based on course concepts

M06 | 8 marks

View Answer

For R(A,B,C), F={AB->C,C->B}, find the keys and BCNF status. Analyse R1(C,B),R2(A,C) for losslessness, BCNF of components and dependency preservation.

Original practice based on course concepts

M07 | 5 marks

View Answer

Jane owns Employee(empID,name,dept,salary). Joe may see only names and salaries for Accounts employees and may not modify data. Give a view-based solution and two conditions needed for enforcement.

Original practice based on course concepts

M08 | 5 marks

View Answer

A Flask search route passes form column into an f-string SQL identifier and binds a separate term. (a) Explain vulnerability and repair. (b) Why is an HTML dropdown insufficient? (c) Give one additional control limiting harm.

Original practice based on course concepts

Part V. Canvas answers and explanations

Published answers and your prior selection appear only in this section. Explanations are independently written. Source issues remain visible rather than being silently corrected.

Week 02 - Weekly Quiz - Answers

W02-Q01 | Published key: D

Back to Question

Published correct option D

Assignment 1 (Conceptual Model) only

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

This is the historical assessment rule tested in the completed Week 2 quiz. The course introduction identifies Assignment 1 as group based and Assignment 2 as individual. It is retained for completeness rather than treated as a database-theory principle.

Study source: L1A pp.29-32.

Open the Canvas review (requires course access).

W02-Q02 | Published key: C

Back to Question

Published correct option C

Each quiz is worth 3%; there are more than 7 quizzes but the total from quizzes is capped at 21%

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published response says that weekly quizzes contribute 3% each and their combined contribution is capped at 21%. This is not the same as assuming there are exactly seven quizzes, or applying the rule to the separate Early Feedback Quiz. This answer records the course information tested at that time.

Study source: L1A pp.23-24.

Open the Canvas review (requires course access).

W02-Q03 | Published key: A

Back to Question

Published correct option A

You may attend in person, or watch online at the same time, or watch the recording later

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The permitted lecture participation modes in the published answer include in-person attendance, watching the stream and watching a recording. This question concerns the course arrangements, not a DBMS property.

Study source: L1A p.8.

Open the Canvas review (requires course access).

W02-Q04 | Published key: A

Back to Question

Published correct option A

Assignment 1 (Conceptual Design)

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published answer identifies Assignment 1. The introduction distinguishes its late-submission arrangement from weekly quizzes and the SQL quiz. Retain this as historical quiz content; actual individual assessment arrangements require their own instructions.

Study source: L1A pp.29,35.

Open the Canvas review (requires course access).

W02-Q05 | Published key: A

Back to Question

Published correct option A

Daily

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The selected published answer is Daily. This is course administration rather than a technical principle.

Study source: Canvas published Week 2 review; L1A general expectations.

Open the Canvas review (requires course access).

W02-Q06 | Published key: C

Back to Question

Published correct option C

The salary for a particular employee

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A particular employee's salary is the data being described. The fact that salary is recorded, its provenance and rules about its handling are metadata about that data. Metadata is a semantic role and can itself be stored as ordinary table content.

Study source: L1B pp.13,36.

Open the Canvas review (requires course access).

W02-Q07 | Published key: C

Back to Question

Published correct option C

Write the code of an application used by endusers to produce a particular report

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

An organisation-specific end-user report is normally produced by its application programmers. Vendor staff can write the DBMS, tuning tools and provide training or support. Read NOT carefully: writing the DBMS itself is a vendor activity.

Study source: L1B pp.29-33.

Open the Canvas review (requires course access).

W02-Q08 | Published key: C

Back to Question

Published correct option C

A database is a collection of data, while a database management system is a software package through which many programs can access a shared database

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The database is the persistent collection of facts; the DBMS is the software providing access and management. Neither the number of tables nor whether data is shared defines a simple database/DBMS boundary.

Study source: L1B pp.4,9-10.

Open the Canvas review (requires course access).

W02-Q09 | Published key: D

Back to Question

Published correct option D

the language used to access the data held in a dbms, either to update the data or to retrieve some data

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

DML supplies operations to retrieve or modify stored data. DDL defines structures and constraints. The question asks for the language category, not the interpreter logic that implements it.

Study source: L1B p.21; L2B pp.4-5.

Open the Canvas review (requires course access).

W02-Q10 | Published key: A

Back to Question

Published correct option A

SQL

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

SQL is the query language. PostgreSQL is a DBMS platform that accepts SQL; Python is a possible host language for an application using a database API.

Study source: L1B pp.16-24.

Open the Canvas review (requires course access).

Week 03 - Early Feedback Quiz - Answers

W03-Q01 | Published key: B

Back to Question

Published correct option B

3%; this cannot be made up using other weekly quizzes

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The Early Feedback Quiz is a separate 3% assessment in the published course information. Its result is not made replaceable by earning more weekly-quiz points.

Study source: L1A pp.23-24.

Open the Canvas review (requires course access).

W03-Q02 | Published key: B

Back to Question

Published correct option B

Show every combination that pairs a SuppID from a row of the Supplier table where the Name is 'Heinz', with the ProdID of a row from the Product table

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published key describes the Cartesian product after filtering Supplier rows to Heinz: there is no equality joining the supplier to its products. Technical note: SuppID exists in both listed tables, so the unqualified SELECT SuppID is ambiguous in executable SQL. Qualifying Supplier.SuppID would express the intended keyed answer; adding the join equality would answer a different query.

Study source: L2B pp.22-24.

Open the Canvas review (requires course access).

W03-Q03 | Published key: B

Back to Question

Published correct option B

SELECT Descr
FROM Product
WHERE SuppID = 523

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Selecting only Descr omits ProdID and SuppID, so it does not return all Product information. The other listed forms select every Product column for the given supplier ID.

Study source: L2B pp.7-13.

Open the Canvas review (requires course access).

W03-Q04 | Published key: B

Back to Question

Published correct option B

SELECT COUNT(DISTINCT StudentID)
FROM AssignmentMark;

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

COUNT(DISTINCT StudentID) counts different students with at least one record. COUNT(*) would count assessment records; the other displayed placements of DISTINCT are not the requested aggregate syntax.

Study source: L2B pp.31-34.

Open the Canvas review (requires course access).

W03-Q05 | Published key: A

Back to Question

Published correct option A

Every row of the table has a different combination of values in this set of columns

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Underlining a collection of columns identifies a key: the combination distinguishes rows. It does not require each component to be unique by itself, to contain ID in its name or to use an integer datatype. In SQL, a declared primary key also makes its components non-NULL.

Study source: L2A pp.7-10; L3A pp.16-20.

Open the Canvas review (requires course access).

W03-Q06 | Published key: D

Back to Question

Published correct option D

an entity type with an attribute which is the identifier of some other entity type

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The unit requires associations with other entity types to be shown as relationship types in the conceptual model. Copying another entity's identifier as an attribute is a relational implementation choice. Composite identifiers, multivalued attributes and relationship attributes are all permitted conceptual constructs.

Study source: L3B p.23; L5 p.7.

Open the Canvas review (requires course access).

W03-Q07 | Published key: C

Back to Question

Published correct option C

A boat can have zero, one, or more jetty values

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The double ellipse around jetty denotes a multivalued attribute. The published answer permits zero, one or several values; the two ellipse outlines do not mean exactly two values.

Study source: L2A p.27; original Boat image.

Open the Canvas review (requires course access).

W03-Q08 | Published key: A

Back to Question

Published correct option A

Whenever a lecturer can teaches several classes, they can't be in the same semester of the same year.

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The diagram makes each Class participate in exactly one teaches relationship, but does not restrict one Lecturer to one Class per semester-year. Class identity includes course, year and semester; that composite key does not impose a timetable constraint on the Lecturer. The published incorrect statement invents such a restriction.

Study source: L2A pp.40-43; original Class ER image.

Open the Canvas review (requires course access).

W03-Q09 | Published key: C

Back to Question

Published correct option C

Dr Polash

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Dr Polash is the published answer for the specified semester. It is retained as historical quiz content rather than a claim about future coordination.

Study source: L1A title and teaching-team pages; Canvas Week 3 review.

Open the Canvas review (requires course access).

W03-Q10 | Published key: C

Back to Question

Published correct option C

a DBMS is cheap, while a file system is more expensive (both in fees for the software, and wages for expert staff)

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A DBMS provides management benefits, but is not guaranteed to be cheaper than direct file storage in software fees and staffing. The other options describe shared control, central constraints and finer-grained access checks.

Study source: L1B pp.40-49.

Open the Canvas review (requires course access).

Week 05 - Weekly Quiz - Answers

W05-Q01 | Published key: A

Back to Question

Published correct option A

CREATE TABLE TableA (
column1 VARCHAR(4),
column2 VARCHAR(8)
);

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Each column declaration gives its name followed by a datatype, separated from the next declaration by a comma. VARCHAR defines a character string column. SQL does not use a colon between a column name and its type in CREATE TABLE.

Study source: L3A pp.6-9.

Open the Canvas review (requires course access).

W05-Q02 | Published key: B

Back to Question

Published correct option B

Any non-NULL value in the depot_name column of AreaDepot, also occurs in the name column of Depot

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The FK requires each non-NULL AreaDepot.depot_name to match Depot.name. It neither requires every Depot to be referenced nor stores a physical row number. The child value set may be a proper subset of the parent key values.

Study source: L3A pp.21-25.

Open the Canvas review (requires course access).

W05-Q03 | Published key: D

Back to Question

Published correct option D

INSERT
INTO Supplier
VALUES (557, 'ACME', '0411203579');

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

INSERT INTO Supplier VALUES (...) adds a row. UPDATE changes existing rows; ADD INTO and EXTEND are not the displayed SQL insertion syntax. Explicit column lists are preferable when writing a new query.

Study source: L3A pp.34-35.

Open the Canvas review (requires course access).

W05-Q04 | Published key: D

Back to Question

Published correct option D

A relationship type T between entities U and V, where T has attributes col3 and col4. The relationship T is many-to-many

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The two copied participant keys implement an association; col3 and col4 are its descriptive attributes. A composite PK(col1,col2), without a further unique constraint on either side, permits many-to-many links. A many-to-one association would need a key or uniqueness restriction on the constrained participant.

Study source: L3B pp.11-15.

Open the Canvas review (requires course access).

W05-Q05 | Published key: A

Back to Question

Published correct option A

2528 BARRACK POINT NSW OAK FLATS DC 103 Clare Runner

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Natural join matches the common postcode. The valid row combines the Barrack Point PostArea tuple for 2528 with Clare Runner's Deliverer tuple for 2528. A person associated only with 3831 cannot be joined to a 2528 row.

Study source: L4A pp.25-26.

Open the Canvas review (requires course access).

W05-Q06 | Published key: B

Back to Question

Published correct option B

Π
r
e
g
n
o

PROJECT[regno](SELECT[year<2002](Car))

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Filter Car using year before projecting regno. Projection first would remove year, leaving the later filter without its required attribute. The other expressions either ask for the wrong years or misuse selection/projection.

Study source: L4A pp.21-24.

Open the Canvas review (requires course access).

W05-Q07 | Published key: C

Back to Question

Published correct option C

```
SELECT uoSCode, count(DISTINCT id)
FROM UoSOffering JOIN AcademicStaff ON instructorId = id
WHERE enrollment >= 200
GROUP BY uoSCode;
```

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Filter offerings with enrollment >= 200, then group by uoSCode and count distinct staff IDs. Repeated offerings by the same instructor should not inflate the number of staff. Grouping by a table name or by instructor answers a different or invalid request.

Study source: L4B pp.4-14.

Open the Canvas review (requires course access).

W05-Q08 | Published key: A

Back to Question

Published correct option A

SELECT c.sid
FROM Catalogue c
WHERE c.pid IN
(SELECT pid
FROM Part
WHERE colour = 'red');

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The IN subquery identifies relevant part IDs, then Catalog links those parts to suppliers. Technical note: the source uses Catalog/color in its schema but Catalogue/colour in its options. The published key captures the intended membership logic; executable SQL must use consistent names, such as Catalog and color. DISTINCT sid can be added if the required output is a set of supplier IDs.

Study source: L4B pp.33-44.

Open the Canvas review (requires course access).

W05-Q09 | Published key: C

Back to Question

Published correct option C

The subquery refer to one of the table variables from the outer query

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A reference to an outer query variable creates correlation. Using the same underlying table twice, aggregation or a WHERE subquery alone does not create the dependency.

Study source: L4B pp.35-38.

Open the Canvas review (requires course access).

W05-Q10 | Published key: A

Back to Question

Published correct option A

find the name of academic staff in the "SCS" department, who taught at least 5 different UoS in 2025

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

For each SCS staff member, the correlated subquery counts distinct unit codes taught by that member in 2025. The outer condition requires at least five. It does not count repeated offerings, return unit codes or require teaching every unit.

Study source: L4B pp.37-44.

Open the Canvas review (requires course access).

Week 06 - Weekly Quiz - Answers

W06-Q01 | Published key: B

Back to Question

Published correct option B

noun

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Nouns suggest things or concepts that may be entity types. Evaluate whether individual instances need identity and information. A noun can instead suggest an attribute, so this is a modelling clue rather than an automatic extraction rule.

Study source: L5 pp.3-4.

Open the Canvas review (requires course access).

W06-Q02 | Published key: B

Back to Question

Published correct option B

Using a verb that connects two nouns

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A verb connecting nouns suggests a relationship, but does not on its own specify how many related instances are permitted or required. Must, may and singular/plural constructions supply cardinality or participation clues.

Study source: L5 pp.4-5.

Open the Canvas review (requires course access).

W06-Q03 | Published key: A

Back to Question

Published correct option A

emp_id | 101 | 102 | 103 | 104

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Mathematical projection returns a set of emp_id values, so repeated 103 appears once. It does not invent 105 and is a valid operation on this relation. SQL without DISTINCT can behave differently.

Study source: L4A pp.11-16,21.

Open the Canvas review (requires course access).

W06-Q04 | Published key: C

Back to Question

Published correct option C

The subquery refer to one of the table variables from the outer query

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The decisive feature is that the inner query refers to an outer variable. This repeats the concept in Week 5 Q9 with a different option order.

Study source: L4B pp.35-38.

Open the Canvas review (requires course access).

W06-Q05 | Published key: D

Back to Question

Published correct option D

barcode, name, cores, cache, speed

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

CPU inherits barcode and name from its superclass, then adds cores, cache and speed. Attributes attached to a sibling subtype are not inherited by CPU.

Study source: L2A pp.49-51; original Computer Part image.

Open the Canvas review (requires course access).

W06-Q06 | Published key: D

Back to Question

Published correct option D

A game is played by at most one competitor

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The arrow from Game toward the relationship diamond constrains each game to at most one relationship instance. It does not impose total participation for Game. The thick line on Competitor requires at least one relationship, not more than one.

Study source: L2A pp.41-43; original Game-Competitor image.

Open the Canvas review (requires course access).

W06-Q07 | Published key: B

Back to Question

Published correct option B

an entity type with an attribute which is the identifier of some other entity type

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

This repeats the conceptual/relational distinction in Week 3 Q6. Represent the association as a relationship, then introduce copied key columns when mapping to relations. A composite identifier is allowed.

Study source: L3B p.23; L5 p.7.

Open the Canvas review (requires course access).

W06-Q08 | Published key: A

Back to Question

Published correct option A

Show every combination that pairs a SuppID from a row of the Supplier table where the Name is 'Heinz', with the ProdID of a row from the Product table

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published key identifies the missing join condition and resulting combinations with Heinz Supplier rows. Technical note: unqualified SuppID is ambiguous because both tables contain it. Qualify the column to make that intended expression executable, without silently adding a join predicate.

Study source: L2B pp.22-24.

Open the Canvas review (requires course access).

W06-Q09 | Published key: C

Back to Question

Published correct option C

SELECT uoSCode, COUNT(DISTINCT year)

FROM UoSOffering

GROUP BY uoSCode

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The output must identify each uoSCode and count its different years, so select both uoSCode and COUNT(DISTINCT year), grouping by uoSCode. A bare aggregate in HAVING is not a Boolean group condition and does not display the count.

Study source: L4B pp.4-14.

Open the Canvas review (requires course access).

W06-Q10 | Published key: B

Back to Question

Published correct option B

DELETE
FROM Supplier;

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

DELETE FROM Supplier removes every row while leaving the table definition. DROP TABLE removes the table itself. DELETE does not use SELECT's star syntax in the displayed command.

Study source: L3A pp.33-37.

Open the Canvas review (requires course access).

Week 07 - Weekly Quiz - Answers

W07-Q01 | Published key: C

Back to Question

Published correct option C

(5, 20, 7, 8, 9)

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The candidate tuple has Y=9, like the existing tuple, but V=20 instead of 6. That violates Y -> V. Sharing only part of UW would not be enough to violate UW -> X; the full determinant must agree.

Study source: L6 pp.17-21.

Open the Canvas review (requires course access).

W07-Q02 | Published key: B

Back to Question

Published correct option B

{A, C, D, E, G, I }

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Starting with AE, E -> D adds D; D -> C adds C; AC -> GI adds G and I. G -> D adds nothing new. No rule yields B, F or H. Thus the closure is ACDEGI.

Study source: L6 pp.31-35.

Open the Canvas review (requires course access).

W07-Q03 | Published key: D

Back to Question

Published correct option D

CE -> D

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

E -> D implies CE -> D by augmentation/reflexive inclusion of C. CE cannot generate A, F or GI: the rule AC -> GI needs A as well as C, and F -> E cannot be reversed.

Study source: L6 pp.26-35.

Open the Canvas review (requires course access).

W07-Q04 | Published key: D

Back to Question

Published correct option D

{A, B, E, F, H}

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

ABEFH contains all mandatory attributes A,B,F,H, which are absent from every RHS. F adds E, E or B adds D, D adds C, and AC adds G,I. Therefore its closure is all nine attributes. It is a superkey even though E is redundant and ABFH is already a candidate key.

Study source: L6 pp.23-35.

Open the Canvas review (requires course access).

W07-Q05 | Published key: A

Back to Question

Published correct option A

R1(E, F) and R2(A, B, C, D, E, G, H, I)

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The first split shares only E. E+ is EDC, which determines neither EF nor the entire second component, so losslessness is not guaranteed. The other splits share B, E and G respectively; B -> D, E -> D and G -> D make the corresponding small component determined by the intersection. Always calculate the actual intersection rather than guessing it from R1 alone.

Study source: L6 binary lossless criterion.

Open the Canvas review (requires course access).

W07-Q06 | Published key: B

Back to Question

Published correct option B

An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The determinant is the full set P: agreement on every P attribute forces agreement on every Q attribute. It does not constrain two rows merely because they agree on some Q attribute, nor does it forbid repeating consistent P/Q combinations in rows that differ elsewhere.

Study source: L6 pp.17-21.

Open the Canvas review (requires course access).

W07-Q07 | Published key: A

Back to Question

Published correct option A

Consider every fd in F except those where the right hand side is a subset of the lefthand side; for each of these fds, check if the left hand side of that fd is a superkey of R. If this is the case for all of the considered fds, then we conclude that R is in BCNF, otherwise R is not in BCNF.

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published rule checks every nontrivial given dependency and requires its determinant to be a superkey. Trivial means RHS is contained in LHS. Checking only one dependency, reversing the subset condition or testing the RHS for superkey status is not the BCNF criterion. For a decomposed schema, also consider its projected implied FDs.

Study source: L6 normal-form sections.

Open the Canvas review (requires course access).

W07-Q08 | Published key: D

Back to Question

Published correct option D

if S is a superkey,then there exists an attribute A in S, such that removing A from S gives a candidate key for R is

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A superkey may already be minimal, or may contain several redundant attributes. It contains a candidate key, but there need not be exactly one removable attribute that yields a key. This is the incorrect universal claim.

Study source: L6 pp.23-25.

Open the Canvas review (requires course access).

W07-Q09 | Published key: A

Back to Question

Published correct option A

BD->AE

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

BD is a candidate key and can determine all attributes, so BD -> AE is compatible with BCNF. B -> D or A -> E would have a nonkey determinant and would also undermine the stated minimal keys. AB -> DE would make AB another superkey and hence introduce an unlisted candidate key under the stated assumptions.

Study source: L6 candidate-key and BCNF sections.

Open the Canvas review (requires course access).

W07-Q10 | Published key: B

Back to Question

Published correct option B

When we join U1 with U2, we get the exactly the contents of the original relation U.

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Losslessness means the natural join of the projected components equals the original relation for every legal instance: no extra combinations and no missing facts. Preserving FDs is a separate property.

Study source: L6 decomposition sections.

Open the Canvas review (requires course access).

Week 08 - Weekly Quiz - Answers

W08-Q01 | Published key: B

Back to Question

Published correct option B

GRANT UPDATE(C) ON T TO X

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

GRANT UPDATE(C) ON T TO X grants modification permission for column C. MODIFY and ALLOW are not the privilege/command keywords in the displayed SQL syntax. This grant concerns a column across rows, not a row-dependent filter.

Study source: L7 pp.14-17.

Open the Canvas review (requires course access).

W08-Q02 | Published key: A

Back to Question

Published correct option A

X can execute SELECT statements that refer to table T, and X can also give SELECT access on T to other users

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The recipient gains SELECT on T and can pass that same privilege on T to others. The grant option does not confer unrelated privileges or rights on other tables.

Study source: L7 pp.14-16.

Open the Canvas review (requires course access).

W08-Q03 | Published key: A

Back to Question

Published correct option A

A user can hold at most 2 roles at any time

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

PostgreSQL roles are not restricted to two memberships per user. A role can be held by many users, and users themselves are represented by roles in the course explanation.

Study source: L7 pp.20-21.

Open the Canvas review (requires course access).

W08-Q04 | Published key: B

Back to Question

Published correct option B

CREATE VIEW V1(B, Howmany) AS
SELECT B, COUNT(A)
FROM R
GROUP BY B;

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

CREATE VIEW ... AS SELECT defines a virtual relation. Group by B, count A and name the output columns B and Howmany. Since A is a primary key and non-NULL, COUNT(A) counts the rows of each group.

Study source: L7 pp.29-31; L4B grouping.

Open the Canvas review (requires course access).

W08-Q05 | Published key: B

Back to Question

Published correct option B

Permit a manager to modify the Salary of any Employee, but not modify other information

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A column-specific UPDATE(Salary) grant expresses this policy without a view. The alternatives require row filtering or aggregate-only access, which a plain column grant does not supply. The complete policy still depends on other privileges not bypassing the restriction.

Study source: L7 pp.17,28-33.

Open the Canvas review (requires course access).

W08-Q06 | Published key: C

Back to Question

Published correct option C

(44,4,5,6,48,49,9)

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The tuple (44,4,5,6,48,49,9) agrees on CD=(5,6) but changes E from 7 to 48, violating CD -> BE. Source note: another option contains a doubled comma and is preserved as printed; it is malformed as a seven-attribute tuple.

Study source: L6 FD definition; Canvas source punctuation.

Open the Canvas review (requires course access).

W08-Q07 | Published key: A

Back to Question

Published correct option A

{C,D,E,G}

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

CDF or its G-based equivalent needs F, because no FD has F on the RHS. CDEG lacks F, so it is not a superkey. Each other option includes D and F plus enough to derive C, then B,E,A and G.

Study source: L6 closure and keys.

Open the Canvas review (requires course access).

W08-Q08 | Published key: A

Back to Question

Published correct option A

R1(A,E) and R2(A,B.C,D,F,G)

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published lossy split shares A, but A+ contains only A. Therefore A does not determine E in R1 or the rest of R2. The other splits share E, C or G; E -> A and C <-> G respectively give the needed small-component determination. The source dot in A,B.C,D is treated as punctuation separating B and C for the intended analysis.

Study source: L6 binary lossless criterion.

Open the Canvas review (requires course access).

W08-Q09 | Published key: A

Back to Question

Published correct option A

An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Equal values on every determinant attribute P force equal values on every dependent attribute Q. Repeated consistent pairs are allowed; the converse Q -> P does not follow.

Study source: L6 pp.17-21.

Open the Canvas review (requires course access).

W08-Q10 | Published key: A

Back to Question

Published correct option A

When we join U1 with U2, we get the exactly the contents of the original relation U.

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The exact original relation is reconstructed by the join. Extra tuples are the hallmark of a lossy association reconstruction; local FD enforcement is a different question.

Study source: L6 decomposition sections.

Open the Canvas review (requires course access).

Week 10 - Weekly Quiz - Answers

W10-Q01 | Published key: B

Back to Question

Published correct option B

A connection represents an ongoing communicationto the database, where setup occurs once (identifying the user, checking password, etc) and then several SQL statements can be performed one after another.

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A connection is an authenticated ongoing DBMS communication context in which several statements can run. A SQL template with placeholders is a parameterised statement, and a result-iteration interface is a cursor.

Study source: L8A pp.19-25.

Open the Canvas review (requires course access).

W10-Q02 | Published key: A

Back to Question

Published correct option A

An HTML document is returned and rendered by the user's local browser

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

In the taught web architecture the application renders an HTML response that the browser displays. Python values and SQL results are intermediate server-side representations; the user does not normally run local Python to print this page.

Study source: L8A pp.5-11.

Open the Canvas review (requires course access).

W10-Q03 | Published key: D

Back to Question

Published correct option D

A cursor points to one row in the result set of a query, and allows the application to process one row after another

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A cursor supports processing query results one row after another through fetch operations. It is not the authenticated user, a table-key lookup that skips SQL, or the definition of a parameterised statement.

Study source: L8A pp.33-37.

Open the Canvas review (requires course access).

W10-Q04 | Published key: A

Back to Question

Published correct option A

A emthod in the file database.py

Visible selection: B | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

database.py constructs the SQL in the supplied scaffold. routes.py receives requests and calls the database layer; templates render the presentation; config.ini holds configuration.

Study source: Lab8 pp.5-7; local scaffold source.

Open the Canvas review (requires course access).

W10-Q05 | Published key: A

Back to Question

Published correct option A

This is a software vulnerability that allows an attacker to submit some SQL fragment as user input which, through insufficiently-checked inclusion of this input within a query string, changes the meaning of a SQL statement that is executed in the dbms, in ways not intended by the application programmer.

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Injection makes attacker-controlled input part of the SQL grammar, changing the query's meaning. Separate value binding prevents that structural interpretation for data values. Error leakage, account compromise and buffer overflow are distinct problems.

Study source: L8B pp.10-16.

Open the Canvas review (requires course access).

W10-Q06 | Published key: D

Back to Question

Published correct option D

read or write of data is much slower than doing so in RAM (main memory)

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Hard-disk access is far slower than memory access and operates through storage blocks rather than independent RAM-like word access. It is persistent and normally cheaper per capacity than RAM in the lecture's comparison.

Study source: L9 pp.10-18.

Open the Canvas review (requires course access).

W10-Q07 | Published key: D

Back to Question

Published correct option D

to hide the slowness of access to secondary storage, by having needed data available in memory for when it will be accessed

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The buffer pool caches pages in RAM so later requests can avoid some slow storage reads. It is a performance mechanism; by itself it neither guarantees durability nor protects logical validity against erroneous applications.

Study source: L9 pp.17-18.

Open the Canvas review (requires course access).

W10-Q08 | Published key: D

Back to Question

Published correct option D

the data is always densely packed, without any gaps

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A heap is unordered by a chosen search key. Inserts can be cheap and a scan can examine its blocks. Deletions and free space mean there is no guarantee of perfectly dense packing.

Study source: L9 pp.20-23.

Open the Canvas review (requires course access).

W10-Q09 | Published key: B

Back to Question

Published correct option B

covering index for the query

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Covering is a property relative to a particular query: the needed values can be obtained from the index without fetching the base records in the course model. Composite, clustered and secondary describe different index properties.

Study source: L9 p.46.

Open the Canvas review (requires course access).

W10-Q10 | Published key: B

Back to Question

Published correct option B

A clustering index on b

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Clustering on b makes equal-b records physically adjacent. In (a,b) order, a is the leading field, so equal-b records can be separated across a groups. A primary-key declaration on a does not change the requested criterion.

Study source: L9 pp.38-43.

Open the Canvas review (requires course access).

Week 11 - Weekly Quiz - Answers

W11-Q01 | Published key: A

Back to Question

Published correct option A

The stored procedure is executed outside the dbms, passing SQL queries to the dbms and receiving the results

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The published false statement puts stored logic outside the DBMS. Stored procedures run inside the database server; host-language application code runs outside and sends requests. Interface note: not every real stored routine returns a whole result set; platform procedures and functions have different result contracts. The key tests execution location.

Study source: L7 pp.63-70; L8A pp.44-46.

Open the Canvas review (requires course access).

W11-Q02 | Published key: A

Back to Question

Published correct option A

The application is a mixture of normal language features, and special syntax for SQL statements

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A call-level interface uses normal host-language calls, passing SQL text and parameters to a library/driver. Special SQL syntax mixed directly into host-language source characterises embedded SQL, not the call-level definition.

Study source: L8A pp.12-13,17-31.

Open the Canvas review (requires course access).

W11-Q03 | Published key: B

Back to Question

Published correct option B

A method in the file database.py

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

database.py is the query-construction layer in the actual local scaffold. This repeats Week 10 Q4 with a different option order.

Study source: Lab8 pp.5-7; local scaffold source.

Open the Canvas review (requires course access).

W11-Q04 | Published key: A

Back to Question

Published correct option A

covering index for the query

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A covering index holds the values required by this particular query. It need not be clustered, and an index being composite alone does not mean it covers every query.

Study source: L9 p.46.

Open the Canvas review (requires course access).

W11-Q05 | Published key: D

Back to Question

Published correct option D

A clustering index on b

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The physical clustering key must lead with b to group the same b values. Leading with a sorts by a first, even if b is included as a second component.

Study source: L9 pp.38-43.

Open the Canvas review (requires course access).

W11-Q06 | Published key: C

Back to Question

Published correct option C

A table scan examines each block of data in the table, and within each block, it checks each record

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A table scan visits the table's data blocks and examines their records, applying any relevant filter. It can work without an index and can process a WHERE clause; it does not directly skip every nonmatching record.

Study source: L10 p.12.

Open the Canvas review (requires course access).

W11-Q07 | Published key: A

Back to Question

Published correct option A

database tuning is done by the system when processing an SQL query, in order to make sure the query gives correct results

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Tuning aims to improve a workload, including index read benefits and update costs. It is not the per-query process that makes arbitrary SQL logically correct. Correct query semantics are required independently of performance tuning.

Study source: L10 pp.35-43.

Open the Canvas review (requires course access).

W11-Q08 | Published key: B

Back to Question

Published correct option B

In query optimisation, the system considers several different ways to execute a particular query, and chooses one which hopefully will run quickly

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The optimiser considers equivalent execution plans and chooses one estimated to be efficient. It is not a vote among repeated executions, a choice of whichever question gives more accurate data, or simply an index-creation operation.

Study source: L10 pp.4-8.

Open the Canvas review (requires course access).

W11-Q09 | Published key: B

Back to Question

Published correct option B

```
SELECT *
FROM T
WHERE 12 < c AND c < 42 AND d = 26;
```

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A hash index on d supports the equality d=26. Candidate rows can then be checked for the c range. A range predicate on d without equality does not get the same ordered-range access from a hash structure.

Study source: L9 p.35; L10 access paths.

Open the Canvas review (requires course access).

W11-Q10 | Published key: B

Back to Question

Published correct option B

```
SELECT a FROM T
WHERE b = 7 AND d = 4;
```

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

In lexicographic (d,c) order, the leading-key equality d=4 provides a useful lookup and b is checked afterwards. The other options constrain a and/or nonleading c without a useful d restriction. The index is not assumed to include a or b automatically.

Study source: L9 pp.42-46.

Open the Canvas review (requires course access).

Week 12 - Weekly Quiz - Answers

W12-Q01 | Published key: C

Back to Question

Published correct option C

Isolation

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

I stands for Isolation. Consistency is C; integrity is an important related concept but is not the I in ACID.

Study source: L11 pp.5,50.

Open the Canvas review (requires course access).

W12-Q02 | Published key: B

Back to Question

Published correct option B

either all the changes made by the transaction are present in the database, or none of the changes are present

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Atomicity is all-or-none visibility of transaction effects. A transaction may contain one or many statements. Preserving business invariants is consistency; reaching the last statement alone does not establish atomicity.

Study source: L11 pp.9-12.

Open the Canvas review (requires course access).

W12-Q03 | Published key: A

Back to Question

Published correct option A

every integrity constraint holds at every state of the database during execution of the transaction code

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Consistency is a valid-state-to-valid-state obligation for a correctly written completed transaction. It does not require every business invariant at every intermediate point, though immediate declared constraints still apply when checked. The commit/rollback statement is a general transaction property rather than the distinctive definition of consistency.

Study source: L11 pp.18-26.

Open the Canvas review (requires course access).

W12-Q04 | Published key: D

Back to Question

Published correct option D

system performance is usually unaffected by the choice of isolation level

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Isolation choices affect concurrency, blocking, retries and which anomalies can occur, so performance is not generally unaffected. The other options describe the SQL model and platform differences presented in the course.

Study source: L11 pp.60-69.

Open the Canvas review (requires course access).

W12-Q05 | Published key: C

Back to Question

Published correct option C

the system ensures that the outcome is the same as in some execution where the transactions are run one after the other

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The DBMS supplies the serial-equivalent execution guarantee for transactions running with the corresponding serializable guarantee. The application still has to write each transaction correctly. Serializable does not mean no committed effects from earlier transactions can ever be seen.

Study source: L11 pp.46-52.

Open the Canvas review (requires course access).

W12-Q06 | Published key: C

Back to Question

Published correct option C

A game is played by at most one competitor

Visible selection: D | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The Game-side arrow means at most one related competitor, while the thick Competitor edge requires participation there. At most one is not exactly one without total participation on Game.

Study source: L2A pp.41-43; original Game-Competitor image.

Open the Canvas review (requires course access).

W12-Q07 | Published key: A

Back to Question

Published correct option A

Whenever a lecturer can teaches several classes, they can't be in the same semester of the same year.

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The invented semester-year limit on a lecturer is not shown. Class's composite key distinguishes classes but does not say that a lecturer cannot teach two different courses in one semester-year. This repeats Week 3 Q8.

Study source: L2A notation; original Class ER image.

Open the Canvas review (requires course access).

W12-Q08 | Published key: D

Back to Question

Published correct option D

barcode, name, cores, cache, speed

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The CPU subtype has its own cores, cache and speed plus inherited barcode and name. Sibling-subtype attributes are not inherited.

Study source: L2A pp.49-51; original Computer Part image.

Open the Canvas review (requires course access).

W12-Q09 | Published key: C

Back to Question

Published correct option C

A table tableW(attrS, attr1, attr2, attr3) where the primary key is (attrS, attr1) and attrS is a foreign key referencing tableS

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A weak entity's identity combines the owner key with its local discriminator. Include all its descriptive attributes too. Owner key alone merges the owner's weak instances; discriminator alone merges instances across owners.

Study source: L3B weak-entity mapping.

Open the Canvas review (requires course access).

W12-Q10 | Published key: B

Back to Question

Published correct option B

Freezer(rack, row, temp);

Sample(barcode, patient_name, freezer_rack, freezer_row, stored_since);

Constraints:

Foreign key Sample(freezer_rack, freezer_row) references Freezer(rack, row)

Visible selection: C | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Each sample is stored in at most one freezer, so copy the freezer's complete composite key and since attribute to Sample. A freezer can contain several samples. The offered join-table alternative omits since and does not enforce the same constrained association. Source note: a complete implementation must also match any total-participation/nullability constraints visible in the original diagram.

Study source: L3B pp.11-15; original sample-storage image.

Open the Canvas review (requires course access).

Week 13 - Weekly Quiz - Answers

W13-Q01 | Published key: B

Back to Question

Published correct option B

OLAP queries often look at a large part of the database, while OLTP queries usually deal with only a few items

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

OLAP commonly scans broad historical data and computes aggregates; OLTP commonly acts on a small set of current operational items. The reversed complexity, update and freshness claims do not describe the course's comparison.

Study source: L12 pp.8-12.

Open the Canvas review (requires course access).

W13-Q02 | Published key: C

Back to Question

Published correct option C

is stored in the contents of tables, and accessed through SQL

Visible selection: A | Score: 0 / 0.3 pts | Differs from the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

ROLAP stores the warehouse data in relational tables and accesses it with SQL. A cube can be a conceptual or visual organisation, not the physical data structure. Special multidimensional physical storage belongs to the contrasting MOLAP approach.

Study source: L12 pp.21,46.

Open the Canvas review (requires course access).

W13-Q03 | Published key: C

Back to Question

Published correct option C

A data warehouse stores data extracted from many operational databases

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

A warehouse commonly integrates extracted data from several operational databases. Its analytical schema need not copy the source schemas, and a visual interface can generate queries instead of every user writing SQL interactively.

Study source: L12 pp.11-19.

Open the Canvas review (requires course access).

W13-Q04 | Published key: C

Back to Question

Published correct option C

"marginal" summary data is included in the output as well as the usual grouped aggregates

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

CUBE retains ordinary grouped aggregates and adds marginal totals and the grand total. With n grouping dimensions it corresponds to 2^n grouping sets, not a cubic running-time claim.

Study source: L12 pp.34-38.

Open the Canvas review (requires course access).

W13-Q05 | Published key: B

Back to Question

Published correct option B

dimensions

Visible selection: B | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Dimensions identify the axes or categories on which measures depend, such as time, product and market. A measure is the value being aggregated; a dimension describes how it is grouped or filtered.

Study source: L12 pp.22-25.

Open the Canvas review (requires course access).

W13-Q06 | Published key: A

Back to Question

Published correct option A

The subset of tuples from PostArea, consisting of the tuples for which state attribute has value 'NSW'

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Selection retains complete tuples whose state equals NSW. It filters rows rather than returning only state or suburb columns, and it does not negate the condition.

Study source: L4A p.22.

Open the Canvas review (requires course access).

W13-Q07 | Published key: A

Back to Question

Published correct option A

Π
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PROJECT[name](SELECT[regno = 'NSW314'](Person JOIN[owner = license] Car))

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Join Person to Car on owner=license, filter the desired regno, then project name. A natural join cannot infer equality between differently named attributes, and projection is not a Boolean selection operator.

Study source: L4A pp.21-26.

Open the Canvas review (requires course access).

W13-Q08 | Published key: C

Back to Question

Published correct option C

An instance of R cannot contain two tuples which have the same value for each attribute in P, and different values for some attribute in Q

Visible selection: C | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The FD restricts pairs that agree on all of P. They must agree on Q. It does not assert its converse or prohibit consistent repetitions within a larger tuple.

Study source: L6 pp.17-21.

Open the Canvas review (requires course access).

W13-Q09 | Published key: D

Back to Question

Published correct option D

(44,4,5,6,48,49,9)

Visible selection: D | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

The keyed tuple agrees on CD=(5,6) but changes E, violating CD -> BE. The doubled-comma option is reproduced from Canvas and is malformed as a seven-attribute tuple; do not infer a new FD from that typo.

Study source: L6 FD definition; Canvas source punctuation.

Open the Canvas review (requires course access).

W13-Q10 | Published key: A

Back to Question

Published correct option A

{C,D,F,G}

Visible selection: A | Score: 0.3 / 0.3 pts | Matches the published key.

Canvas evidence: the option-level review explicitly marks this option as "This was the correct answer."

Canvas publishes A, CDFG. Its closure contains CDFG, then B,E through CD -> BE, then A through E -> A. Technical issue: option C, DEFG, is also a superkey: G -> C gives C, then CD -> BE gives B, then E -> A gives A. Thus this single-choice item has two mathematically valid options under the listed FDs. Retain the published key for provenance and use the closure proof for learning.

Study source: L6 closure and superkey definitions; independent calculation.

Open the Canvas review (requires course access).

Part VI. Written-task solutions

These are independently written worked answers and suggested marking points. Alternative semantically correct solutions are possible. Source-based technical assumptions are stated explicitly.

Revision paper A - SQL and access - Solutions

R-A1 | Worked solution

Back to Question

```
SELECT COUNT(*) AS activityCount
FROM Activity a JOIN Member m ON a.Leader = m.MemberID
WHERE a.Duration > 2 AND m.Role = 'Tester';
```

The count is over activities. MemberID is a key, so the leader join cannot multiply an activity. An unnecessary join to WorksOn could multiply it by its workers.

Supplied revision paper p.1, A(i); independently written solution

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R-A2 | Worked solution

Back to Question

```
SELECT a.ActivityID, COUNT(w.MemberID) AS memberCount
FROM Activity a LEFT JOIN WorksOn w
  ON w.ActivityID = a.ActivityID AND w.NumberDaysWorked > 3
GROUP BY a.ActivityID;
```

This interprets 'for each activity' to include zero counts. The filter belongs in ON; COUNT(w.MemberID) ignores the unmatched NULL-extended row. The composite WorksOn key ensures each member contributes at most once per activity. An inner join reports only activities with a qualifying worker.

Supplied revision paper p.1, A(ii); independently written solution

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R-A3 | Worked solution

Back to Question

```
SELECT COALESCE(SUM(NumberDaysWorked), 0) * 8 AS totalHours
FROM WorksOn WHERE ActivityID = 57;
```

Sum contributed days, not Activity.Duration. COALESCE defines no reported contributions as a zero total. If NULL days mean unknown work, the reporting policy must handle that uncertainty explicitly rather than claiming the work was zero.

Supplied revision paper p.1, A(iii); independently written solution

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R-A4 | Worked solution

Back to Question

```
SELECT a.ActivityID FROM Activity a
WHERE a.ActivityType = 'Testing'
  AND a.Duration = (
    SELECT MAX(b.Duration) FROM Activity b
    WHERE b.ActivityType = 'Testing'
```

```
  )
ORDER BY a.ActivityID;
```

Both the candidate population and the maximum population are filtered to Testing. Equality to MAX preserves every tie; a one-row limit would not.

Supplied revision paper pp.1-2, A(iv); independently written solution

R-A5 | Worked solution

Back to Question

Every supplied non-NULL Leader value must match an existing MemberID. Several activities may share a leader, and a member need not lead an activity. This FK alone does not require a leader, because Leader is nullable. Parent-key changes and deletions are handled according to the declared referential action; none is specified here.

Supplied revision paper p.2, A(v); independently written solution

R-A6 | Worked solution

Back to Question

```
PROJECT_Name(
 SELECT_NumberDaysWorked>3 AND ActivityType='Testing'(
  (Member JOIN_Member.MemberID=WorksOn.MemberID WorksOn)
   JOIN_WorksOn.ActivityID=Activity.ActivityID Activity
 )
)
```

Join on participant identifiers, filter both conditions, then project Name. Mathematical projection eliminates duplicate names; the requested name-only output cannot distinguish different members sharing a name. SQL needs DISTINCT for matching set behaviour.

Supplied revision paper p.2, A(vi); independently written solution

R-A7 | Worked solution

Back to Question

```
CREATE VIEW activity39_members AS
SELECT DISTINCT m.Name, m.Role
FROM Member m JOIN WorksOn w ON m.MemberID = w.MemberID
WHERE w.ActivityID = 39;
GRANT SELECT ON activity39_members TO joe;
```

Assume Joe has connection/schema access but no other privileges bypassing this interface. Grant no modification rights. If direct or role-derived base-table rights exist, those paths must also be restricted: a view does not revoke earlier grants. PostgreSQL may also need schema USAGE for Joe.

Supplied revision paper p.2, A(vii); independently written solution

R-A8 | Worked solution

Back to Question

```
CREATE INDEX member_role_joined ON Member(Role, Joined);
```

The source says Members while its schema declares Member; this solution uses Member. A tree can locate Role='Tester', scan the Joined range within that group, then fetch output values. A larger (Role,Joined,MemberID,Name) index can supply the whole query in the course's covering model, at extra storage and maintenance cost. Benefit depends on selectivity; no measured speedup is claimed.

Supplied revision paper p.2, A(viii); independently written solution

Revision paper B - Airport EER - Solutions

R-B1 | Worked solution

Back to Question

Engineer has a composite name attribute with given and family components. Composite does not mean multivalued or identifying. Its relational mapping can store the two components separately; emp_id remains the entity identifier.

Supplied revision paper p.3, B(i); independently written solution

R-B2 | Worked solution

Back to Question

No. The underlined modelCode identifies one Model instance, which has one maker value. Different Airplane instances can share that Model, but the same modelCode cannot identify two distinct maker-specific Model instances. An alternative composite identifier (maker,modelCode) could permit this, but is not the supplied diagram.

Supplied revision paper p.3, B(ii); independently written solution

R-B3 | Worked solution

Back to Question

```
CREATE TABLE Engineer (
  emp_id INTEGER PRIMARY KEY, given VARCHAR(80), family VARCHAR(80)
);
CREATE TABLE Model (
  modelCode VARCHAR(30) PRIMARY KEY, maker VARCHAR(80)
);
CREATE TABLE Airplane (
  tailCode VARCHAR(30) PRIMARY KEY, name VARCHAR(100),
  modelCode VARCHAR(30) NOT NULL REFERENCES Model(modelCode)
);
CREATE TABLE Certified (
  emp_id INTEGER REFERENCES Engineer(emp_id),
  modelCode VARCHAR(30) REFERENCES Model(modelCode),
  PRIMARY KEY(emp_id,modelCode)
);
CREATE TABLE Airliner (
  modelCode VARCHAR(30) PRIMARY KEY REFERENCES Model(modelCode),
  maxPassengers INTEGER
);
CREATE TABLE CargoFreighter (
  modelCode VARCHAR(30) PRIMARY KEY REFERENCES Model(modelCode),
  maxCargo NUMERIC
);
```

The thick Airplane arrow becomes a mandatory Model FK. Certified is many-to-many. The default ISA is partial and overlapping, so neither total coverage nor disjointness should be invented. Datatypes are reasonable design choices, not source-prescribed constraints.

Supplied revision paper p.3, B(iii); independently written solution

Revision paper C - DoctorsInfo - Solutions

R-C1 | Worked solution

Back to Question

ProvidorNo -> Specialty: existing provider 183 is associated with Lung, while the new row associates the same determinant with ENT. The patient and visit facts shown do not contradict the other supplied dependencies. Different diagnoses alone violate none of the listed FDs.

Supplied revision paper p.4, C(i); independently written solution

R-C2 | Worked solution

Back to Question

Start with VisitID and ProvidorNo. VisitID -> VisitDate,PatientID adds VisitDate and PatientID. PatientID -> Age,PostCode,City adds those three. ProvidorNo -> Specialty adds Specialty. PostCode -> City adds nothing new. Diagnosis remains absent, so the closure is all attributes except Diagnosis. The pair is not a superkey under the given FDs.

Supplied revision paper p.4, C(ii); independently written solution

R-C3 | Worked solution

Back to Question

PostCode -> City is nontrivial and PostCode+={PostCode,City}, which is not the whole relation. Its determinant is therefore not a superkey. One such violation disproves BCNF. Merely pointing to repeated data is not the formal test.

Supplied revision paper p.4, C(iii); independently written solution

R-C4 | Worked solution

Back to Question

Use Provider(ProvidorNo,Specialty), PK(ProvidorNo), FD ProvidorNo -> Specialty. This component is BCNF.

Use VisitDetail(VisitID,VisitDate,PatientID,Age,City,PostCode,ProvidorNo,Diagnosis). Its projected FDs include VisitID -> VisitDate,PatientID; PatientID -> Age,PostCode,City; PostCode -> City, with their implications. A candidate key is (VisitID,ProvidorNo,Diagnosis): these three cannot be generated from others, and together generate the entire component. VisitDetail is not BCNF, for example because PostCode -> City still has a nonkey determinant. The prompt asks for two relations, not a completed BCNF normalisation.

Supplied revision paper p.4, C(iv); independently written solution

R-C5 | Worked solution

Back to Question

For the split in R-C4, the intersection is ProvidorNo. ProvidorNo -> Specialty makes the intersection determine all of Provider. The binary lossless-decomposition criterion therefore holds for every legal original instance. A successful join of only the sample data would not prove this universal property.

Supplied revision paper p.4, C(v); independently written solution

R-C6 | Worked solution

Back to Question

Yes, for R-C4. ProvidorNo -> Specialty is enforceable locally in Provider. All the other listed FDs are local to VisitDetail. The union of these projected dependencies contains, and therefore implies, the original FD set. This argument is separate from proving reconstruction without spurious tuples.

Supplied revision paper p.5, C(vi); independently written solution

Revision paper D - Flask security - Solutions

R-D1 | Worked solution

Back to Question

Start with routes.py, whose handlers receive path and form data. In search_users_byname, follow request.form['searchfield'] and ['searchterm'] into database.search_users_customfilter. In update and delete handlers, follow userid and editable values to their database helpers. Templates describe normal browser input, but do not constrain a crafted request.

Inspect database.py at every SQL execution site, including dictfetchall and dictfetchone wrappers. Trace SQL text and parameter values to their origin. search_users_customfilter interpolates attributename and filtertype but binds filterval; enforce server-side allowlists for structural choices. list_users_equifilter also interpolates attributename, whose safety depends on each caller. update_single_user and delete_user interpolate userid and should bind it as data. check_login and add_user_insert illustrate fixed SQL with separately supplied values, subject to separate authentication and privilege concerns.

Use fixed SQL structure, trusted identifier/operator mappings and driver value binding. A dropdown alone is insufficient, and an empty second execute argument does not repair already interpolated input. Treat authorisation, credential handling and controlled error reporting as additional concerns. This is inspection of the local ZIP source, not a claim of live exploitation or complete security testing.

Supplied revision paper p.5, D and local Week 8 ZIP; independently written solution

Additional skill drills - Solutions

D01 | Worked solution

Back to Question

The rows change the instance. The index changes physical design and related metadata, but not the logical Student attributes or intended answer. Adding a column instead would change the logical schema. Neither operation supplies an output-order guarantee.

Original practice - Foundations; see the corresponding guide chapter

D02 | Worked solution

Back to Question

Shared data reduces inconsistent duplicated addresses and centralises capabilities such as constraints and access checks. It creates coordination, administration and availability requirements, including software and staffing costs. A DBMS assists management but does not automatically make every application or value correct.

Original practice - Foundations; see the corresponding guide chapter

D03 | Worked solution

Back to Question

Chef to Prepares is a thin arrow toward the diamond: zero or one participation. Dish to Prepares is a thick plain line: at least one and possibly several. Do not reverse the arrow merely because Dish is casually called the one side.

Original practice - EER; see the corresponding guide chapter

D04 | Worked solution

Back to Question

CPU has barcode, name, cores and speed, but does not inherit RAM.size. Overlap permits one Product in both subtypes. Partial covering permits a Product in neither subtype. Membership overlap and coverage are independent restrictions.

Original practice - EER; see the corresponding guide chapter

D05 | Worked solution

Back to Question

Room(buildingID,roomNo,capacity), PK(buildingID,roomNo), FK(buildingID) to Building. The PK makes both components non-NULL. This captures weak identity and mandatory ownership, but does not require each Building to own at least one Room.

Original practice - Mapping; see the corresponding guide chapter

D06 | Worked solution

Back to Question

Add nullable spaceID to Employee, referencing ParkingSpace(spaceID), with UNIQUE(spaceID). The FK enforces existence and uniqueness prevents sharing a supplied space. A space can remain unused; an employee can have no assigned space. Do not create two mandatory circular FKs.

Original practice - Mapping; see the corresponding guide chapter

D07 | Worked solution

Back to Question

The result is (1,1),(2,0). LEFT JOIN on ActivityID with days>3 in ON, group by the Activity ID and count the child's MemberID. Filtering days in WHERE removes activity 2; COUNT(*) counts its NULL-extended row as one.

Original practice - SQL; see the corresponding guide chapter

D08 | Worked solution

Back to Question

The results are 4, 3, 2, 40 and 40/3. NULL is excluded from value aggregates but its row contributes to COUNT(*). DISTINCT changes the values supplied to the indicated aggregate; it is not the syntax COUNT DISTINCT(score).

Original practice - SQL; see the corresponding guide chapter

D09 | Worked solution

Back to Question

Both students: there is no required unit missing for either candidate. If the business rule also demands at least one actual enrolment or a nonempty requirement set, add that separate condition deliberately. Division with candidates drawn only from Enrolled has a different domain.

Original practice - Subqueries; see the corresponding guide chapter

D10 | Worked solution

Back to Question

2=1 is false and 2=NULL is unknown, so 2 IN the set is unknown and its negation stays unknown. WHERE rejects it. Use a correlated NOT EXISTS with the intended matching equality, or exclude NULL if that matches the requested semantics.

Original practice - Subqueries; see the corresponding guide chapter

D11 | Worked solution

Back to Question

The projection discards year before the selection needs it. Use PROJECT_regno(SELECT_year<2002(Car)). Selection retains rows; projection retains attributes and removes duplicate tuples in the set algebra. Read nested expressions inside out.

Original practice - Algebra; see the corresponding guide chapter

D12 | Worked solution

Back to Question

Join selected red Parts to Catalog and project sid. Repeat for green; intersect the sid sets, then join to Suppliers and project name. A single part need not have both colours. Intersecting names can incorrectly identify distinct suppliers who happen to share a name.

Original practice - Algebra; see the corresponding guide chapter

D13 | Worked solution

Back to Question

A and D cannot be generated and must be in every key. (AD)+=ABCD and neither singleton works, so AD is the sole candidate key. A->B violates BCNF because A+=ABC misses D; B->C also violates it. A composite key alone is not the definition of a violation.

Original practice - Design theory; see the corresponding guide chapter

D14 | Worked solution

Back to Question

AB/AC shares A, which determines AB, so it is lossless. AB/BC shares B, which determines neither whole component, so losslessness is not guaranteed. Merely placing all attributes somewhere does not preserve the original associations.

Original practice - Design theory; see the corresponding guide chapter

D15 | Worked solution

Back to Question

Create a department/AVG(salary) view and grant SELECT on it while excluding base-table access through all privilege paths. Small groups and differences between overlapping reports can reveal individual information, so evaluate the actual released output. A view is a mechanism, not an automatic privacy proof.

Original practice - Security; see the corresponding guide chapter

D16 | Worked solution

Back to Question

Dependent grants through X may be removed according to revoke behaviour, but Y's independent owner-granted path can remain. Check every source of the privilege rather than equating one revocation with removal of all access. A SELECT grant never authorises unrelated UPDATE rights.

Original practice - Security; see the corresponding guide chapter

D17 | Worked solution

Back to Question

No. The identifier changes SQL structure and is not protected by binding the separate term. Map permitted choices to fixed server-side identifiers and reject unknown choices, while keeping values parameterised. A data placeholder cannot generally represent an arbitrary column name.

Original practice - Applications; see the corresponding guide chapter

D18 | Worked solution

Back to Question

Injection changes SQL interpretation through untrusted text. Stored XSS executes attacker content when later rendered by a browser. Bound SQL values address grammar/data separation; output escaping and controlled HTML handling address browser interpretation. Safely storing a string does not make it safe executable HTML.

Original practice - Applications; see the corresponding guide chapter

D19 | Worked solution

Back to Question

(role,joined) supports equality on the leading key and then a date range. With joined first, the date range can span many roles, which are filtered afterwards. Actual benefit depends on selectivity, storage and maintenance costs; describe the access path instead of promising a particular speedup.

Original practice - Indexes; see the corresponding guide chapter

D20 | Worked solution

Back to Question

Index lookup and pointer processing add overhead, while scattered fetches access most of the table anyway. A sequential scan may read those pages more efficiently. Cache state and placement affect the result, so an index is an alternative access path rather than an unconditional improvement.

Original practice - Indexes; see the corresponding guide chapter

D21 | Worked solution

Back to Question

The business transfer is split into two atomic units and lacks all-or-none effects as one task. A committed debit is not undone by rolling back the later credit. Group the logically connected updates into one appropriate transaction with whole-task failure handling.

Original practice - Transactions; see the corresponding guide chapter

D22 | Worked solution

Back to Question

This is write skew: different writes depend on overlapping observations and jointly violate the at-least-one-on-call rule. Protect the shared invariant with appropriate transaction coordination or a serializable guarantee and correct failure handling. Separate row locks chosen without considering the rule may not protect it.

Original practice - Transactions; see the corresponding guide chapter

D23 | Worked solution

Back to Question

The groupings are (market,product), (market), (product), and (). The empty grouping gives the grand total. Sales amount is a measure; market and product identify dimensions. Marginal totals supplement, rather than replace, the ordinary groups.

Original practice - Analytics; see the corresponding guide chapter

D24 | Worked solution

Back to Question

Technical access is capability, not sufficient authority for every use. Consider Collective Benefit, Authority to Control, Responsibility and Ethics. Identify relevant rights, governance decisions, intended benefits and accountability. This is a course-concept answer, not a jurisdiction-specific legal conclusion.

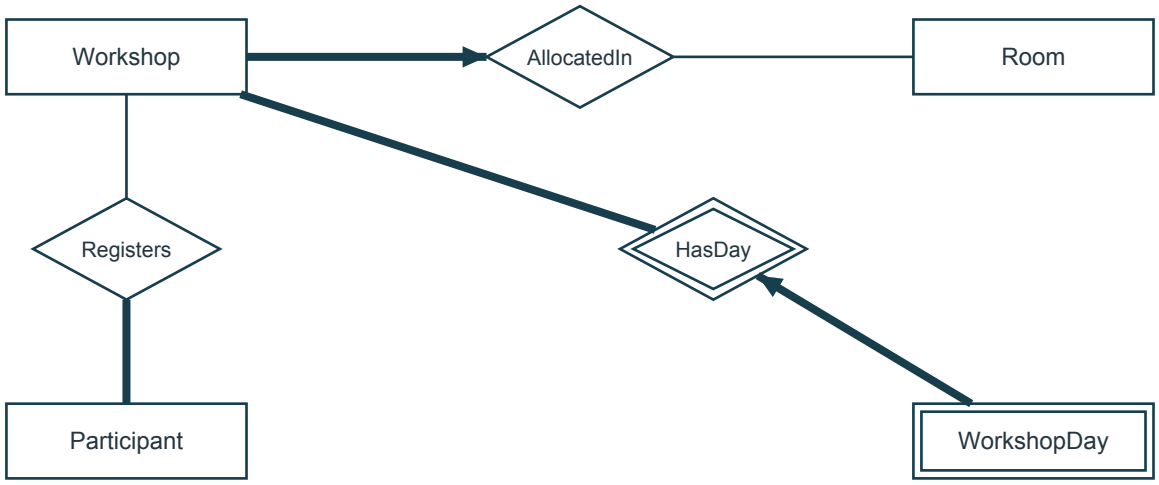
Original practice - Governance; see the corresponding guide chapter

Practice exam - 50 marks - Solutions

M01 | Worked solution

Back to Question

High-level EER: attributes and identifiers are specified in the accompanying text.



Arrowheads face the relationship diamond. Thick edges require participation.
High-level model: include the attributes and identifiers stated in the solution.

Use Workshop, Room and Participant entities with the given identifiers. Capacity is a Room attribute. AllocatedIn has a thick arrow from Workshop into its diamond, and a thin plain Room edge unless another constraint is stated. Registers is many-to-many, with a thick plain Participant edge and thin plain Workshop edge.

Use weak WorkshopDay with local discriminator date and attribute chairperson. Draw the double rectangle, double identifying diamond and dashed discriminator underline. WorkshopDay's identifying participation is a thick arrow toward the diamond; Workshop's HasDay edge is thick/plain because this prompt requires at least one day.

No uniqueness of workshop or participant names is implied. The chairperson can remain a recorded name because no identified Person is required. Suggested marking: entity identities/attributes (2), room link (2), registration (2), day/chair identity and notation (2).

Original practice based on course concepts

M02 | Worked solution

Back to Question

For (a), use R-A2's left join, put NumberDaysWorked>3 in ON, group by a.ActivityID and count w.MemberID. For (b), use R-A4's equality to MAX over Testing activities, retaining the same outer category filter and ORDER BY ActivityID.

Suggested marking: all-activity preservation (1), correct filter placement (1), count/group including zero (1); filtered maximum (1), ties (1), output attribute and ordering (1). Semantically equivalent solutions are acceptable.

Original practice based on course concepts

M03 | Worked solution

Back to Question

```
CREATE TABLE Building (
  buildingID INTEGER PRIMARY KEY, name VARCHAR(80)
);
CREATE TABLE Room (
  buildingID INTEGER REFERENCES Building(buildingID),
  roomNo INTEGER, capacity INTEGER,
  PRIMARY KEY(buildingID,roomNo)
);
CREATE TABLE Customer (
  customerID INTEGER PRIMARY KEY, name VARCHAR(80)
);
CREATE TABLE Booking (
  bookingID INTEGER PRIMARY KEY, startTime TIMESTAMP,
  customerID INTEGER NOT NULL REFERENCES Customer(customerID),
  buildingID INTEGER NOT NULL, roomNo INTEGER NOT NULL,
  FOREIGN KEY(buildingID,roomNo) REFERENCES Room(buildingID,roomNo)
);
```

The room columns jointly reference a Room. Two separate FKs do not enforce the same composite pair. Do not add uniqueness to Booking's room FK because several bookings are allowed. Suggested marking: entity tables/keys (2), weak-room identity and ownership (2), customer constraint (2), composite Room FK and mandatory participation (2).

Original practice based on course concepts

M04 | Worked solution

Back to Question

(a) PROJECT_name(SELECT_year<2002(Person JOIN_owner=license Car)). Keep the join attributes until the join; set projection eliminates duplicate names.

(b) Covering is relative to a query's required values. Clustering concerns physical locality of base records. Either property may occur without the other.

(c) No: lexicographic (a,b) groups by a first, so a b value can occur in separated a groups. A physical clustering order led by b addresses that requirement. Suggested marking: two marks per part, including the reason.

Original practice based on course concepts

M05 | Worked solution

Back to Question

Start CE; add D by C->D, then A by DE->A, then B by A->BC. CE+=ABCDE. E is absent from every RHS and is mandatory in a key. AE, CE and DE each generate all attributes; removing either component makes each closure incomplete. These are all keys: E and BE cannot generate A,C or D, so any key needs E and at least one of A,C,D.

Suggested marking: correct closure steps (2); all three candidate keys, minimality and completeness (2).

Original practice based on course concepts

M06 | Worked solution

Back to Question

AB and AC are the candidate keys: each closes to ABC, while no singleton does. C->B is nontrivial and C+=BC, so R is not BCNF.

The split shares C, which determines CB, proving losslessness. R1 is BCNF with C as key; R2 has no projected nontrivial FD with a nonkey determinant. The components' local dependencies do not imply AB->C: AB cannot generate C. Thus the lossless BCNF split is not dependency-preserving.

Suggested marking: keys/minimality (2), violation (2), losslessness/component BCNF (2), dependency-preservation argument (2).

Original practice based on course concepts

M07 | Worked solution

Back to Question

```
CREATE VIEW accounts_pay AS
SELECT name, salary FROM Employee WHERE dept = 'Accounts';
GRANT SELECT ON accounts_pay TO joe;
```

Joe must have no direct or role-derived path bypassing the filter or permitting modification. The owner must have relevant underlying privileges, and Joe needs connection/schema access to use the view. Do not grant modification privileges. A view does not identify separate application users sharing one DBMS account.

Suggested marking: correct rows/columns (2), privilege (1), two enforcement conditions (2).

Original practice based on course concepts

M08 | Worked solution

Back to Question

(a) column changes SQL structure and is not protected by binding term. Map allowed public choices to fixed server-side identifiers and reject unknown choices. Keep term bound by the driver; a data placeholder does not stand for a column name.

(b) A crafted request can bypass the normal browser widget, so server-side enforcement is required.

(c) Limit the application's DBMS privileges and unnecessary sensitive data; apply per-user authorisation. Controlled errors and protected credentials complement these controls but do not replace fixing injection.

Suggested marking: vulnerability (1), structural allowlist and value binding (2), client-side limit (1), relevant additional protection (1).

Original practice based on course concepts

Sources and use limits

Canvas: ISYS2120 Data and Information Management, course 73755; closed quiz reviews captured 14 November 2026. No new attempt was started or submitted. The visible attempt order supplies the option labels. Five source images are embedded, including their repeated uses.

Local sources: all 18 lecture/review PDFs listed in the companion guide; tutorial sheets for Weeks 2-8 and 10-13; ISYS2120 revision questions-2026.pdf; and Tut/isys2120_airline_webapp_2025.zip. Page references use physical PDF pages.

Equations use readable operator transcriptions. Question wording, spelling and substantive distractors follow the visible source; source punctuation problems are flagged in the explanations. Revision-paper task wording is lightly reformatted. The written solutions, drills and practice exam are study material, not official answers or an exam prediction.

The source code was inspected locally. No claim is made of a live university application test, complete exploit audit, unseen Canvas question variants, or independent verification of personal exam arrangements.

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